Corrective Overoptimism

Brandon S.C. Chan*
Yale University

First version: March 2026
This version: October 2026
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Abstract

Booms in infrastructure investment often coincide with investor exuberance and losses, yet the infrastructure remains long after these episodes. I show that investor overoptimism can correct for a general class of market failure: agents with diagnostic expectations over-extrapolate good news and conduct irreversible investments, pushing what was once a market failure toward the efficient level. This mechanism – corrective overoptimism – is embedded in a dynamic spatial model with endogenous infrastructure and estimated on data from England and Wales, 1700–1881, spanning the canal and railway booms. Corrective overoptimism can be a potent channel for growth.

JEL Classification: D84, E22, E32, N73, O18, O33, R12, R42.

Keywords: Corrective overoptimism, diagnostic expectations, infrastructure investment, spatial development, technology diffusion, Industrial Revolution.

*I am grateful to Costas Arkolakis, Joel Flynn and Charles Hodgson for their indispensable guidance. I also thank Bill Janeway, Victor Li, Justin Yifu Lin, Michael B. Wong and Michael Yao for useful feedback, and participants at the Cleveland Fed and the 25th Carroll Round at Georgetown University for helpful comments. Special thanks also to the Cambridge Group for the History of Population and Social Structure (Campop) for providing data. This research was supported by the Yale Center for Research Computing and the Friedman Family Research Fellowship for urban development. Online appendices and replication code are available at https://github.com/brandonscchan/corrective-overoptimism.

brandon.chan@aya.yale.edu

> Wherever you are in the country, you are just a short distance from the great arterial system. At any moment, you could integrate with this great life-force traversing England; become an essential part of the nation or retreat to the suburban domesticity that you call home.
>
> — Simon Foxell, *Mapping England*

The mass expansion of railways, the electric grid, and the fibre network each coincided with dramatic boom–bust cycles: the Railway Mania of 1846, the Great Depression of 1929, and the Dot-Com Bubble of 2000.[1] Investors suffered enormous losses. Yet from each episode emerged permanent connective infrastructure that reshaped the economy. Janeway (2018) and Morck (2022) argue that this pattern is not coincidental: exuberance-driven technology buildout has been central to the growth process. I formalize this argument and quantify its importance.

The core idea is simple. A monopolist who must sink capital into infrastructure before demand is realized will generically underinvest, because market power drives a wedge between the private return and the social return. If that monopolist over-extrapolates good news – forming beliefs that are too optimistic after positive signals – the resulting over-investment can close the gap. And because the capital is irreversible, the correction survives even after beliefs revert. I call this mechanism *corrective overoptimism*.

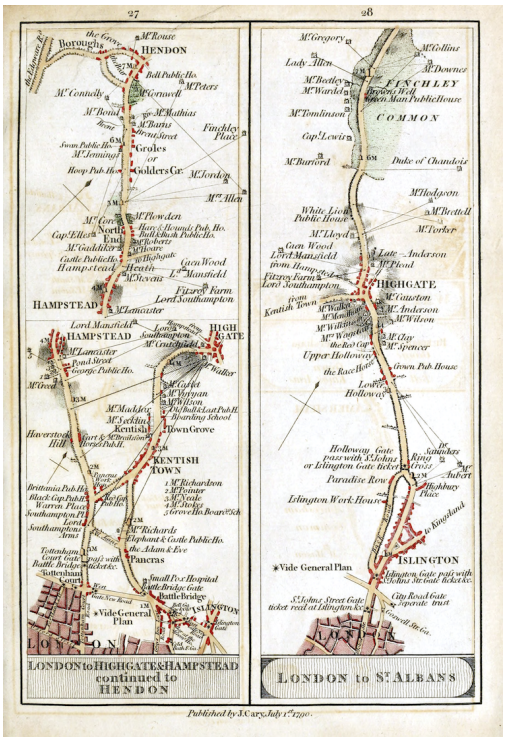
The idea that financial exuberance can be productive has a long history (Keynes, 1936; Kindleberger, 1978; Shiller, 2000). A large literature shows that bubbles can relieve financial constraints, deploying capital more efficiently and crowding in investment for innovation (see, e.g., Guerron-Quintana et al., 2023; Hirano and Yanagawa, 2016; Martin and Ventura, 2012; Miao and Wang, 2014; Nanda and Rhodes-Kropf, 2017). My contribution is to show that behavioral bias can correct not just for financial frictions, but for a broad class of market failures rooted in externalities (Lipsey and Lancaster, 1956). The mechanism I identify operates even in a world without funding constraints: what matters is that an agent invests above their rational choice in a market with positive externalities, and that the resulting investment is sunk.

I model the bias using diagnostic expectations (DE), introduced by Bordalo et al. (2018) as a formalization of the representativeness heuristic of Kahneman and Tversky (1972). Under DE, agents overweight the likelihood of states that have become more likely in light of recent news. A growing literature has shown DE to improve our understanding of credit cycles, asset prices, and business cycle dynamics (Bianchi et al., 2024; Bordalo et al., 2026, 2019, 2020; Elekdag et al., 2025; Gennaioli et al., 2016; L'Huillier et al., 2023; Maxted, 2023).[2] I exploit the interaction between the overreactive property of DE and irreversible investment. When a diagnostic infrastructure monopolist receives good news about future demand, it overestimates the return to building and invests beyond the rational level. Once the capital is sunk, the economy benefits permanently.

The simplest form of market failure is a static monopoly. I show corrective overoptimism can close this static gap in Section I. Externalities and spillovers then bloom into many richer forms, particularly through space and time (Bresnahan and Trajtenberg, 1995; Lucas, 1988; Romer, 1990). The rise of the railway coincides with the collapse of the cost of letter delivery and migration. Telegraphs and fibre networks allow us to exchange ideas at new rates (Berkes et al., 2026; Cinnirella

[1] See, for example, White (1990), Pástor and Veronesi (2009), Odlyzko (2010), Greenstein (2015), and Caoui and Steck (2025).

[2] In complementary work, Flynn and Sastry (2025) and Flynn et al. (2026) study contagious narratives and persistence learning over the business cycle, while Bizzarri and d'Arienzo (2024) analyze the social value of information overreaction in abstract coordination settings.



Great North Road out of London, 1790, John Cary. Courtesy of Duke University via HathiTrust.

et al., 2024; Hanlon, 2025; Mokyr, 2005). Each connective revolution increased the ability for each individual to stand on others' shoulders. A single railway company or local turnpike operator cannot capture all the positive benefit they bring to innovation and communication on the other side of the nation. The accumulation of these innovations in generating hockey-stick growth means that corrective overoptimism can accidentally kickstart a wave of innovations through spillovers. I contextualize this idea of spillovers to the structural transformation of England and Wales in 1700–1881 in Section II – documenting with local projections and shift-share instruments that regional construction surges systematically reversed below trend and compressed subsequent railway operating yields (Arnold and McCartney, 2011; Campbell, 2012; Jordà, 2005; Odlyzko, 2010) – and develop the model in Section III. Section IV then assesses the quantitative significance of corrective overoptimism in the growth process.

A distinguished literature has studied the relationship between transport infrastructure and economic outcomes. Fogel (1964) and Fishlow (1965) asked what the American economy would have looked like without the railway. Donaldson and Hornbeck (2016) brought this question into a market-access framework (see also Ahlfeldt et al., 2015; Balboni, 2025; Donaldson, 2018; Heblich et al., 2020; Hodgson, 2018; Nagy, 2023; Redding and Turner, 2015). Most recently, Hornbeck and Rotemberg (2024) showed, by accounting for input distortions (Baqaee and Farhi, 2020; Hsieh and Klenow, 2009; Restuccia and Rogerson, 2008), that the aggregate productivity gains from American railways were far larger than the earlier estimates of Fogel (1964) and Donaldson and Hornbeck (2016) suggested, because infrastructure has outsized effects in distorted economies. Fajgelbaum and Schaal (2020), Allen and Arkolakis (2022), and Allen et al. (2025) characterize optimal infrastructure networks in static environments, while Brancaccio et al. (2020), Bordeu (2025), and Jiang (2026) study decentralized transport provision. In the dynamic spatial literature, Kleinman et al. (2023) incorporate forward-looking capital accumulation and migration into a dynamic spatial

general equilibrium, and Allen and Donaldson (2020) develop a theory of path-dependent spatial economies through agglomeration.

The counterfactual tradition running from Fogel to Donaldson and Hornbeck has produced sharp answers to the question of what infrastructure is worth once built. A complementary question is whether the right amount gets built in the first place, and whether the psychological biases of the investors who finance it might, against the usual intuition, improve the outcome. I take up this question. Doing so requires a model in which the network emerges endogenously from forward-looking investment decisions that are subject to systematic bias, and a structural approach to evaluate counterfactuals we cannot observe directly.

I build on the spatial development framework of Desmet and Rossi-Hansberg (2014) and Trew (2020), contributing four parsimonious extensions to the rational baseline before introducing diagnostic expectations. First, I introduce an endogenous diffusion of ideas (Cai et al., 2022; Nelson and Phelps, 1966). This allows me to capture the dynamic gains from endogenous infrastructure investment through innovation diffusion and idea exchange, something which is not captured from just reducing trade costs and increasing market access (Proposition 3). Second, I incorporate local manufacturing operating wedges (Desmet et al., 2020; Hornbeck and Rotemberg, 2024) alongside competitive county distribution and haulage services (Keibek, 2017; Shaw-Taylor and Wrigley, 2014), capturing the allocative efficiency gains as transport expansion shifts labor out of agriculture into wedge-distorted industry. Third, I incorporate forward-looking county infrastructure companies with local toll-setting power. Drawing on the historical record of turnpike trusts, canal companies, and early railways – each promoted under Parliamentary acts to build durable route capacity while competitive carriers operated freight (Bogart, 2014; Ward, 1974) – I show that the counties' dynamic Markov perfect investment problem reduces to a tractable Tobin marginal-q criterion (Hayashi, 1982) with a unique equilibrium under own-cost diagonal dominance (Propositions 4 and 7). Fourth, I show that the framework of Desmet and Rossi-Hansberg (2014) for firms and workers extends to diagnostic expectations while preserving identical within-period choices (Propositions 1, 2, 5, and 6).

The main analytic insight is that diagnostic expectations can systematically counteract the underinvestment generated by decentralized infrastructure provision, and that the effect compounds over time. A county infrastructure company's marginal-q criterion evaluates only the operating-service rent gain within its own jurisdiction, whereas three general-equilibrium spillover channels – through-trade price reductions, labor reallocation into wedge-distorted manufacturing, and accelerated spatial knowledge diffusion – deliver gains across the wider economy that no local promoter can internalize. Because diagnostic expectations amplify the variance of perceived continuation values around positive news while installed route capacity is irreversible, the extra network construction triggered during booms is permanent, whereas the investment shortfalls from amplified bad news are merely deferred to the next expansion.

Taking the model to 1710–1881 data on output, relative industrial prices, occupational shares, spatial sorting, and multimode transport capacity, the Simulated Method of Moments estimates the diagnosticity parameter at $\widehat{\theta} = 0.739$ (SE $= 0.090$), rejecting rational expectations ($\theta = 0$, $t = 8.21$, $p < 0.0001$) and matching the $\theta \approx 0.74$–0.91 range documented for credit spreads, earnings forecasts, and firm investment (Bordalo et al., 2018, 2026, 2019). Re-estimating the model under rational expectations ($\theta = 0$) matches the secular trends in output and employment, but cannot simultaneously generate the 1830–1850 railway-mania surge, the post-1850 deceleration, and the negative serial correlation and spatial reversals of infrastructure additions. Out of sample, the diagnostic model also tracks the untargeted post-1820 acceleration of aggregate real wages (3.11 times the 1710 level by 1881 versus 3.28 in the data), whereas the re-estimated rational model stalls at 1.72 (Allen, 2009). A belief-neutralization counterfactual (R | DE) that sets $\theta = 0$ while

holding all other estimated parameters, initial stocks, project opportunities, and primitive shocks fixed shows that diagnostic overoptimism expanded the 1881 British transport network by +43.79 percent (+54.47 percent in the Midlands–North corridor), raised real output per worker by +20.96 percent, and lifted worker real wages by +12.97 percent. Larger installed route capacity accelerated structural transformation out of agriculture by −12.9 percentage points, raised manufacturing frontier productivity by +10.90 percent, and lowered the relative price of manufactures by −13.79 percent (Table 7 and Figure 9; compare Balboni, 2025; Donaldson, 2018; Donaldson and Hornbeck, 2016; Fajgelbaum and Redding, 2022; Fogel, 1964; Trew, 2020). Decomposing the +20.96 percent output gain across the model's three spillover channels shows that direct transit savings account for 25.9 percent, labor reallocation into wedge-distorted manufacturing accounts for 29.5 percent, and faster spatial knowledge diffusion accounts for 44.7 percent.

Moving forward, this paper has three goals. First, to provide a clear illustration of how diagnostic expectations can correct for market failure. Second, to achieve accurate estimates of a model with overoptimism, disciplined by historical context. And third, to assess the empirical relevance of corrective overoptimism. As each goal builds on the last, so do the following sections. Section I provides a simple example. Section II lays out the historical setting. Section III develops the full model, with proofs in Appendix A. Section IV presents the quantitative analysis. Section V concludes.

I A Simple Example

Consider a two-period regional economy that produces wheat and exports it through a monopoly railway. Exporters' willingness to pay for shipping is governed by the world wheat price Z_t, which follows a stationary AR(1) process $Z_{t+1} = \rho Z_t + \varepsilon_{t+1}$ with innovation ε_{t+1} and persistence $\rho \in (0, 1)$. Inverse shipping demand is linear,

$$P_t(Q_t) = Z_t - bQ_t, \qquad b > 0, \tag{1}$$

where Q_t is the volume shipped. To carry wheat at date $t + 1$, the railway must sink irreversible capacity $K_{t+1} \geq 0$ at date t at unit construction cost $c > 0$, and it sells the installed capacity forward to exporters at the market-clearing price $P_{t+1}(K_{t+1})$, so that shipments equal capacity, $Q_{t+1} = K_{t+1}$. Operating costs are zero, and innovations are bounded below, $\varepsilon_{t+1} \geq -c$, which keeps the clearing price nonnegative for every capacity up to the first-best level K^{SP}_{t+1} derived below.[3]

A First-Best Capacity

Expected Marshallian social surplus at date t from installing capacity K_{t+1} is $\mathbb{E}_t[S_{t+1}(K_{t+1})] = \mathbb{E}_t\left[\int_0^{K_{t+1}} (Z_{t+1} - bq)\, dq\right] - cK_{t+1} = (\mathbb{E}_t[Z_{t+1}] - c)K_{t+1} - \frac{1}{2}bK^2_{t+1}$. Equating expected marginal social benefit to unit construction cost yields first-best capacity:

$$\mathbb{E}_t\left[P_{t+1}(K^{\text{SP}}_{t+1})\right] = c \quad \Longrightarrow \quad K^{\text{SP}}_{t+1} = \frac{\mathbb{E}_t[Z_{t+1}] - c}{b}. \tag{2}$$

[3] Forward sale is what makes the first-order conditions below exact. If the owner could instead leave capacity idle at zero cost, it would ship $\min\{K_{t+1}, Z_{t+1}/(2b)\}$ ex post, the planner's and owner's conditions would become $\mathbb{E}_t[(P_{t+1}(K))_+] = c$ and $\mathbb{E}^\theta_t[(\text{MR}_{t+1}(K))_+] = c$, and the closed forms below would be lower bounds on the optimal capacities. Online Appendix S.A analyzes ex-post quantity competition and route entry.

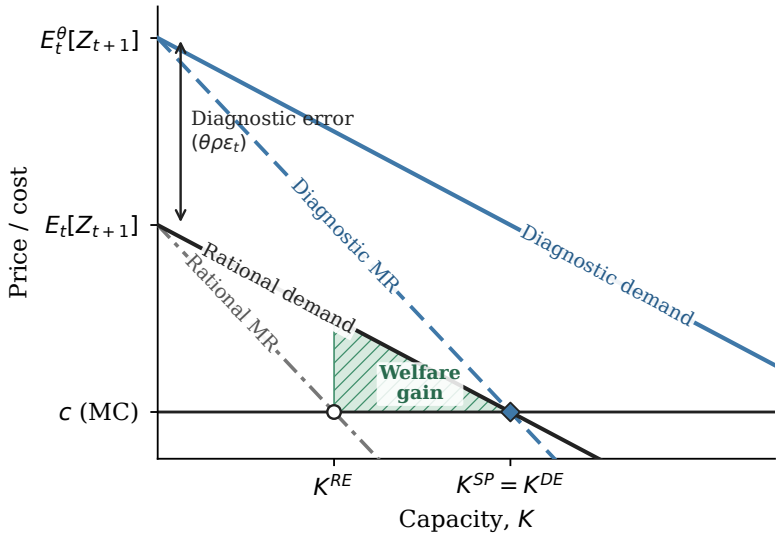


Figure 1: Railways for Wheat Exports

Notes: Linear inverse shipping demand $P = Z - bQ$ and marginal revenue $\text{MR} = Z - 2bQ$. Equating expected marginal revenue rather than marginal social benefit to unit capacity cost c leads rational expectations ($\theta = 0$) to build half the socially efficient capacity ($K^{\text{RE}} = \frac{1}{2}K^{\text{SP}}$), whereas diagnostic overreaction to positive news ($\theta > 0$, $\varepsilon_t > 0$) shifts perceived demand outward and expands sunk capacity toward K^{SP}.

B Monopolist Investment under Rational and Diagnostic Expectations

Following Bordalo et al. (2018), the promoter's diagnostic expectation of Z_{t+1} overweights current news $\mathbb{E}_t[Z_{t+1}] - \mathbb{E}_{t-1}[Z_{t+1}] = \rho\varepsilon_t$:

$$\mathbb{E}_t^\theta[Z_{t+1}] = \mathbb{E}_t[Z_{t+1}] + \theta \underbrace{\left(\mathbb{E}_t[Z_{t+1}] - \mathbb{E}_{t-1}[Z_{t+1}]\right)}_{\text{news}} = \mathbb{E}_t[Z_{t+1}] + \theta\rho\varepsilon_t, \qquad \theta \geq 0, \tag{3}$$

nesting rational expectations at $\theta = 0$. Because the monopolist cannot price discriminate across inframarginal shippers, it equates perceived marginal revenue $\mathbb{E}_t^\theta[\text{MR}_{t+1}] = \mathbb{E}_t^\theta[Z_{t+1}] - 2bK_{t+1}$ to unit cost c:

$$K_{t+1}^{\text{DE}} = \frac{\mathbb{E}_t^\theta[Z_{t+1}] - c}{2b} = \frac{\mathbb{E}_t[Z_{t+1}] + \theta\rho\varepsilon_t - c}{2b}. \tag{4}$$

Under rational expectations ($\theta = 0$), the appropriation wedge between marginal social benefit $Z - bK$ and marginal revenue $Z - 2bK$ halves investment:

$$K_{t+1}^{\text{RE}} = \frac{\mathbb{E}_t[Z_{t+1}] - c}{2b} = \frac{1}{2}K_{t+1}^{\text{SP}}. \tag{5}$$

After positive news $\varepsilon_t > 0$, diagnostic optimism expands sunk capacity by $K_{t+1}^{\text{DE}} - K_{t+1}^{\text{RE}} = \theta\rho\varepsilon_t/(2b) > 0$ and changes expected physical social surplus relative to the rational monopoly benchmark by

$$\mathbb{E}_t\left[S_{t+1}(K_{t+1}^{\text{DE}}) - S_{t+1}(K_{t+1}^{\text{RE}})\right] = \frac{\theta\rho\varepsilon_t}{8b}\left[2(\mathbb{E}_t[Z_{t+1}] - c) - \theta\rho\varepsilon_t\right]. \tag{6}$$

Equation (6) captures the second-best logic of Lipsey and Lancaster (1956) for any market failure in which expected marginal social benefit exceeds expected marginal private benefit by an appropriation or externality wedge $\Delta(K) > 0$ at the rational outcome K_{t+1}^{RE}: by the envelope theorem,

a small diagnostic expansion has zero first-order private loss to the investor while generating a strictly positive first-order social gain $\Delta(K_{t+1}^{\mathrm{RE}})\,\partial K_{t+1}^{\mathrm{DE}}/\partial(\theta\rho\varepsilon_t) > 0$ that narrows the underinvestment wedge $\mathbb{E}_t[Z_{t+1}] - c$, whereas second-order in $(\theta\rho\varepsilon_t)^2$ it introduces a convex overshooting cost once capacity exceeds K_{t+1}^{SP}. Every positive shock satisfying $0 < \theta\rho\varepsilon_t < 2(\mathbb{E}_t[Z_{t+1}] - c)$ strictly raises expected social welfare above the rational monopoly outcome—attaining the first best at $\theta\rho\varepsilon_t = \mathbb{E}_t[Z_{t+1}] - c$—and because capacity is irreversible (Arrow, 1968; Dixit and Pindyck, 1994), the capital built during the boom persists after beliefs revert.

II Historical Setting and Motivating Evidence

Britain's Industrial Revolution combines three empirical regularities across England and Wales over 1710–1881 that motivate our dynamic spatial framework. First, aggregate growth reallocated labor out of agriculture from a substantial pre-existing industrial base, sorting manufacturing toward the Midlands and Northern coalfields while expanding distribution and transport employment in every county (Broadberry et al., 2014; Crafts and Harley, 1992; Shaw-Taylor and Wrigley, 2014). Second, turnpikes, canals, and railways were promoted by local committees under Parliamentary acts, separating durable route construction from competitive carriage alongside persistent local manufacturing wedges (Bogart, 2014; Hornbeck and Rotemberg, 2024; Trew, 2020; Ward, 1974). Third, major waves of infrastructure building surged after favorable fundamental news and systematically overshot—producing sharp boom–bust reversals in capacity additions and post-boom compression in operating yields (Arnold and McCartney, 2011; Campbell, 2012; Odlyzko, 2010).

A Spatial Structural Transformation and Distribution Labor

To measure regional structural change on a consistent geography, I place county occupational series on fixed 1831 ancient-county boundaries across twelve benchmark dates from 1710 to 1881, combining Keibek (2017) with the CAMPOP national benchmarks (Shaw-Taylor et al., 2019), London registration-county series (Field and Shaw-Taylor, 2019), and the county population weights of Wrigley (2011) (Online Appendix S.B). Occupations follow the Primary–Secondary–Tertiary (PST) classification of Wrigley (2010): primary covers agriculture, forestry, and fishing; secondary groups mining with manufacturing and construction because coalfield geography parallels industrial production (Allen, 2009; Clark and Jacks, 2007); and tertiary comprises transport, communications, dealers, and sellers, isolating the distribution and haulage activities required to move goods across space.

Figures 2 and 3 highlight two features of this transition. Nationally, structural transformation proceeded from a large eighteenth-century industrial base: secondary employment reached half of included employment by 1817, while between 1851 and 1881 the primary share fell from 31 to 19 percent, the secondary share rose from 56 to 62 percent, and the tertiary share expanded from 13 to 18 percent as growing trade volumes required expanding distribution services (Herrendorf et al., 2012; Shaw-Taylor and Wrigley, 2014). Regionally, primary employment remained concentrated in the South and East and secondary industry sorted toward the Midlands and Northern coalfields, whereas distribution and transport employment expanded in every county while remaining thickest around London and the northern industrial hubs (Bogart et al., 2022; Heblich et al., 2020).

B Decentralized Infrastructure Provision, Carriage, and Local Distortions

Unlike the state-planned trunk systems of continental Europe (Milward and Saul, 1973; Smith, 1990), British turnpikes, canals, and railways were promoted decentralizedly by regional committees

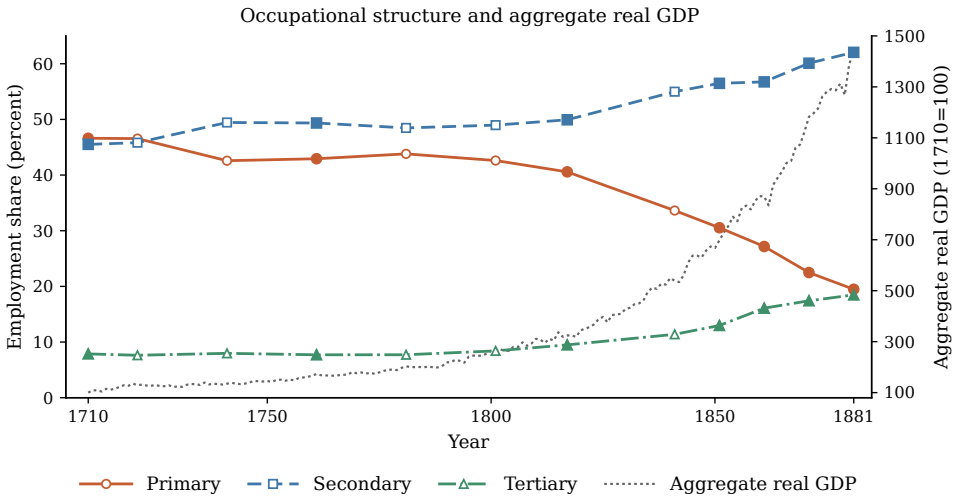


Figure 2: Occupational Structure and Aggregate Real GDP

Notes: Adult male employment shares across primary (excluding mining), secondary (manufacturing, mining, construction), and tertiary (dealers, sellers, transport, communications) sectors: filled markers denote CAMPOP benchmarks (Shaw-Taylor et al., 2019) and open markers denote aligned Keibek (2017) estimates. Right axis: Great Britain real GDP (1710 = 100; Broadberry et al. 2022; Feinstein 1972).

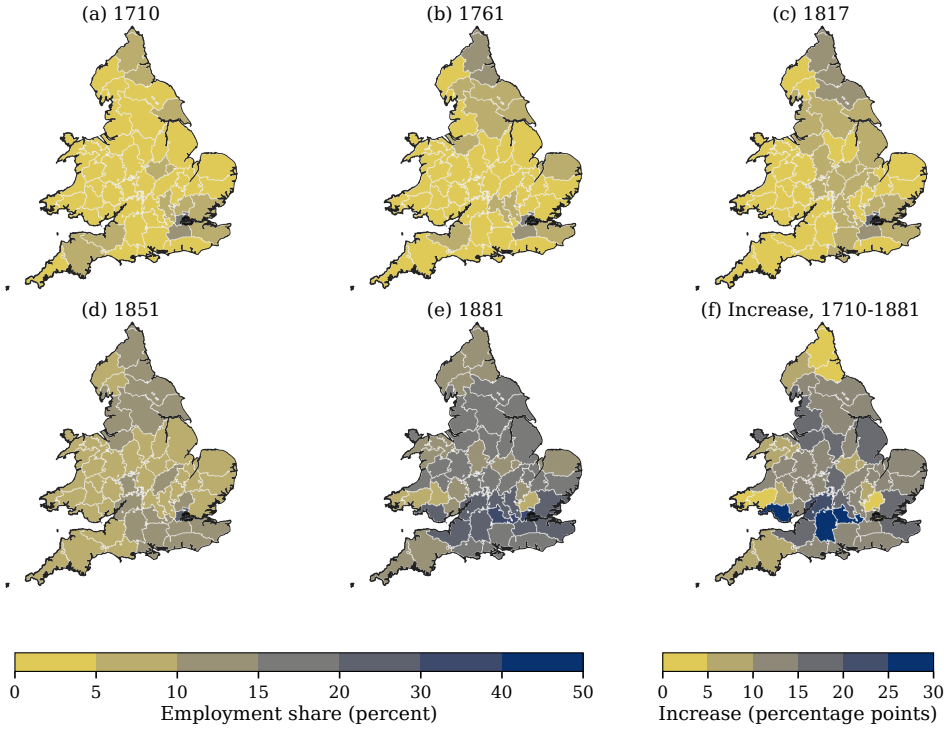


Figure 3: Spatial Expansion of Distribution and Transport Employment, 1710–1881

Notes: Panels (a)–(e) plot county tertiary employment shares at five benchmark dates; Panel (f) plots the 1710–1881 percentage-point increase. County shares through 1851 combine Keibek (2017) and Shaw-Taylor et al. (2019) on 1831 ancient-county boundaries, and 1881 projects the reconstructed latitude-band surface (Online Appendix S.B); the black outline marks London (Field and Shaw-Taylor, 2019).

whose completed links reshaped market access and technology diffusion (Berkes et al., 2026; Bogart, 2014; Mokyr, 2005; Trew, 2020). Across all three eras, statutory organization separated durable route provision from competitive carriage: turnpike trusts, canal companies, and early railway promoters raised capital to build route capacity and collect tolls within parliamentary-authorized corridors, while independent carriers hired operating labor and vehicles to move freight (Hadfield, 1981; Turnbull, 1987; Ward, 1974).

Network expansion also interacted with persistent local distortions in manufacturing. Desmet, Greif and Parente (2020) document how interregional competition eroded local craft resistance to mechanization unevenly across regions, and Hornbeck and Rotemberg (2024) show that transport improvements generate first-order productivity gains when they reallocate inputs toward wedge-distorted manufacturing producers. Figure 4 plots the cumulative multimode transport stock and annual physical additions over 1710–1881, illustrating the clustered construction waves of the 1790s canal mania and the 1830s–1840s railway booms.

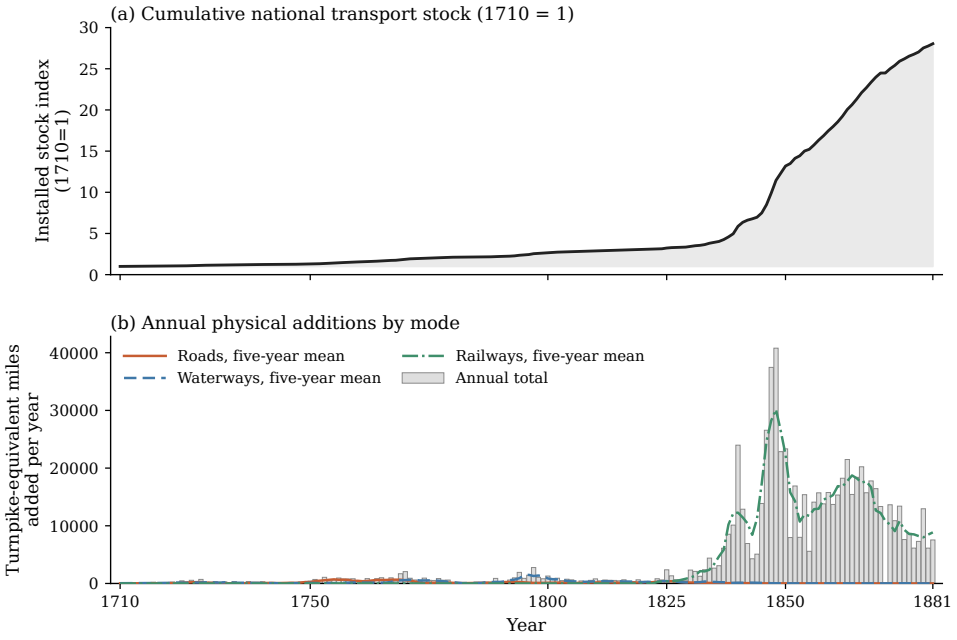


Figure 4: Transport Stock and Installed Additions, 1710–1881

Notes: Turnpikes, waterways, and railways aggregated into turnpike-equivalent miles (road+10×waterway+ 50 × railway). Top panel: cumulative national stock (1710 = 1). Bottom panel: annual physical additions (bars) and five-year moving averages by mode (lines).

C Boom–Bust Reversals and Yield Compression in Infrastructure

Historical accounts of British canal and railway promotion emphasize recurrent cycles of promoter optimism and subsequent disappointment. During the 1830s and the 1845–1848 Railway Mania, prospectuses extrapolated from early trunk successes to advertise 10–15 percent dividends even as realized equity returns averaged under 5 percent and half of Mania-authorized mileage went unbuilt (Odlyzko, 2010). Campbell (2012) shows that weekly railway share prices overreacted to short-run dividend news, while Arnold and McCartney (2011) and Shaen (1847) document systematic cost overruns and post-opening yield shortfalls across early canal and railway companies.

To test whether infrastructure building adjusted smoothly to fundamentals or overshot during

booms, let $r^I_{b,t} = (T_{b,t+1} - T_{b,t})/T_{b,t}$ denote the annual growth rate of transport capacity in latitude band b and $c^I_{b,t} = \mathrm{HP}_{6.25}[\log(10^{-3} + r^I_{b,t})]$ its Ravn and Uhlig (2002) cyclical component. Following Jordà (2005) and Bordalo et al. (2018, 2020), I estimate local projections across horizons $h \in \{1, \ldots, 5\}$ for future regional capacity additions $c^I_{b,t+h}$ and aggregate British railway gross operating yields R_{t+h}/K_{t+h} (working receipts over paid-up capital from Mitchell 1988):

$$c^I_{b,t+h} = \alpha_{bh} + \delta_{th} + \rho_h c^I_{b,t} + u_{b,t+h}, \qquad \frac{R_{t+h}}{K_{t+h}} = \alpha^R_h + \beta^R_h \Delta \ln K_t + u^R_{t+h}. \tag{7}$$

Under rational construction smoothing, a building surge at date t predicts positive future additions above trend ($\rho_h > 0$). Under boom-time overshooting, excess capacity depresses subsequent additions below trend ($\rho_h < 0$) and compresses subsequent operating yields ($\beta^R_h < 0$). Estimating ρ_h directly on the future level $c^I_{b,t+h}$ places no $-c^I_{b,t}$ term on the left-hand side, avoiding mechanical differencing (Jordà, 2005; López-Salido et al., 2017).

Table 1: Local Projections of Infrastructure Investment and Railway Yields

	(1) $h=1$	(2) $h=2$	(3) $h=3$	(4) $h=4$	(5) $h=5$
Panel A. Dependent variable: Regional investment cycle $c^I_{b,t+h}$ (25 latitude bands, 1710–1880)					
Investment cycle $c^I_{b,t}$ (TWFE)	−0.221	−0.215	−0.121	−0.028	−0.011
	(0.019)	(0.020)	(0.020)	(0.020)	(0.019)
Investment cycle $c^I_{b,t}$ (2SLS)	−0.273	−0.202	−0.084	−0.088	−0.004
	(0.054)	(0.057)	(0.054)	(0.056)	(0.028)
Band and year fixed effects	Yes	Yes	Yes	Yes	Yes
First-stage F-statistic (2SLS)	175.6	174.8	174.4	173.2	172.3
Observations	4,250	4,225	4,200	4,175	4,150
Panel B. Dependent variable: Railway gross operating yield R_{t+h}/K_{t+h} (percent, 1843–1881)					
Log paid-up capital growth $\Delta \ln K_t$ (OLS)	−13.00	−13.71	−13.65	−12.49	−11.15
	(2.23)	(2.06)	(2.31)	(2.37)	(2.19)
R^2	0.46	0.52	0.51	0.45	0.40
Observations	38	37	36	35	34

Notes: Local projections (7) at horizons $h = 1, \ldots, 5$ years (Jordà, 2005). Panel A estimates regional regressions across 25 latitude bands (1710–1880) by TWFE and by 2SLS using shift-share instruments $Z_{b,t} = (e^{\mathrm{coal}}_b \bar{c}^{I,\mathrm{coal}}_{-b,t},\ \pi_{M,b,1710}\bar{c}^{I,M}_{-b,t},\ (0.5 + e^{\mathrm{coal}}_b)\bar{c}^I_{-b,t})'$ that interact predetermined GIS coalfield exposure e^{coal}_b (Bogart et al., 2022) and 1710 manufacturing shares $\pi_{M,b,1710}$ with leave-one-out regional addition cycles; band-clustered standard errors are in parentheses. Panel B regresses British railway gross working receipts over paid-up capital R_{t+h}/K_{t+h} (percent) on annual log paid-up capital growth $\Delta \ln K_t$ over 1843–1881 (Mitchell, 1988), with Newey–West HAC standard errors in parentheses.

Table 1 documents sharp post-boom reversals in both physical construction and realized operating yields. In Panel A, a one-unit cyclical surge in regional capacity additions is followed by below-trend additions over the next three years ($\rho_1 = -0.221$, SE $= 0.019$; $\rho_2 = -0.215$, SE $= 0.020$; $\rho_3 = -0.121$, SE $= 0.020$). Instrumenting regional building with predetermined coalfield and 1710 manufacturing exposure interacted with leave-one-out regional addition cycles (first-stage $F > 172$) yields closely matching reversals ($\rho^{\mathrm{IV}}_1 = -0.273$, SE $= 0.054$; $\rho^{\mathrm{IV}}_2 = -0.202$, SE $= 0.057$), confirming that fundamental-driven booms overshoot and contract.[4] In Panel B, railway paid-up capital surges $\Delta \ln K_t$ predict lower subsequent gross operating yields across all five horizons ($\beta^R_1 = -13.00$

[4]Online Appendix S.B (Table S.B.1) shows that these reversals are robust to two-way clustering by latitude band and year, leave-corridor-out ($|k - b| \geq 3$) shift-share instruments, one-sided backward-looking detrending, untransformed level and inverse-hyperbolic-sine addition rates, unfiltered cumulative changes, and separate pre-railway (1710–1829) and railway (1830–1880) subsamples.

to $\beta_5^R = -11.15$, $R^2 = 0.40$–0.52), consistent with the 34 percent shortfall between prospectus and realized net traffic yields documented in Shaen (1847)'s parliamentary audit.

Taken together, medium-horizon construction reversals ($\rho_h < 0$), predictable post-boom yield compression ($\beta_h^R < 0$), and systematic shortfalls relative to prospectus forecasts discriminate against standard rational mechanisms. First, under multi-period time-to-build and convex adjustment costs (Hayashi, 1982; Kalouptsidi, 2014; Kydland and Prescott, 1982), multi-year gestation lags generate positive short-run completion momentum ($\rho_1 = +0.067$ during the railway era and $\rho_1^{1s} = +0.159$ under one-sided detrending; Online Appendix Table S.B.1), but additions return to trend once projects finish rather than dropping below trend at medium horizons ($\rho_2, \rho_3 < 0$). Nor can known construction lags explain why date-t capital growth $\Delta \ln K_t \in \mathcal{F}_t$ predicts five years of operating-yield compression in Panel B or why realized yields systematically missed promoters' own parliamentary prospectuses (Shaen, 1847). Second, under Bayesian learning and irreversible real options (Arrow, 1968; Bernanke, 1983; Dixit and Pindyck, 1994; Jovanovic, 1982; Pástor and Veronesi, 2009), uncertainty raises the option value of waiting—delaying and smoothing entry rather than bunching it into booms—while martingale belief updating implies gradual continuation ($\rho_h > 0$) and unpredictable ex post return errors conditional on date-t investment. Third, in the aggregate time series, cyclical additions load strongly on current and lagged sectoral output cycles ($b_0 = 2.95$, $b_1 = 1.58$) but have zero predictive power for next-year output ($b_{\text{lead}} = -0.005$, SE $= 0.004$), indicating that promoters extrapolated recent industrial expansions rather than acting on private foresight of future fundamentals (Cochrane, 1991). Finally, year fixed effects δ_{th} in Panel A absorb aggregate interest-rate and risk-premium fluctuations so that the 2SLS reversals reflect cross-regional overreaction to real industrial news within each year, while Mania-era prospectuses advertised 10–15 percent dividends—above, not below, historical norms (Campbell, 2012; Odlyzko, 2010).

III The Model

A The Spatial Economy

Time is discrete, $t \in \{0, 1, 2, \dots\}$. Building on the continuous-space geography of Lucas and Rossi-Hansberg (2002) and Rossi-Hansberg (2005) and its extension to spatial growth and endogenous transport by Desmet and Rossi-Hansberg (2014) and Trew (2020), the economy occupies the unit interval $\ell \in [0, 1]$, partitioned into J county jurisdictions $\mathcal{C}_j = [c_{j-1}, c_j]$, with output, employment, and route infrastructure measured as densities per unit of land. Three sectors operate across space: agriculture (A) and manufacturing (M), which produce the two consumption goods, and transport and distribution services (T). Competitive land leasing assigns each site $\ell \in [0, 1]$ to a single goods sector ($e_A(\ell, t) + e_M(\ell, t) = 1$, $e_i(\ell, t) \in \{0, 1\}$), while competitive transport carriers operate alongside goods producers within each county.

A continuum of worker households of mass $\bar{L}_t$ supplies one unit of labor inelastically and holds equal shares in the national portfolio of land leases, producer net profits, carrier capacity rents, and factor-wedge rebates. Under constant participation, England and Wales population P_t^{EW} determines the household mass (normalized to 100 in 1710):

$$\bar{L}_t = 100 \frac{P_t^{EW}}{P_{1710}^{EW}}, \tag{8}$$

with each household receiving $1/\bar{L}_t$ of aggregate nonlabor income. In each county $j \in \{1, \dots, J\}$, a parliamentary-authorized infrastructure company chooses completion intensity x_{jt} to maximize expected discounted land rents inside $\mathcal{C}_j$ net of manufactured construction expenditure.

As in dynamic spatial general equilibrium frameworks (Kleinman et al., 2023), each period t unfolds in two stages conditional on inherited knowledge $A_{it}(\ell)$, route capacity $T_t(\ell)$, and aggregate productivity gaps $\mathbf{z}_t$, separating forward-looking state transitions from within-period spot market clearing:

1. *Pre-discovery commitments:* County companies commit $\mathbf{x}_t \in \mathcal{X}_t$; goods entrants announce pure sector $e_i(\ell, t)$, research probability $\phi_i(\ell, t)$, innovation bill Ψ_i, and hired production labor $L_i(\ell, t)$ at a fixed local contract wage $w(\ell, t)$; carriers commit operating labor $L_T(\ell, t)$ at $w(\ell, t)$; and households choose residence density $n(\ell, t)$.

2. *Post-discovery spot clearing and transitions:* Local innovation gains $G_{it}(\ell, \omega)$ realize; output utilization $Y_i(\ell, t, \omega)$, county tolls $\tau_j(t, \omega)$, spot prices $p_i(\ell, t, \omega)$, trade flows $H_i(\ell, t, \omega)$, and consumption $c_i(\ell, t, \omega)$ clear markets statewise; and discoveries and completed projects update $A_{i,t+1}(\ell)$ and $T_{t+1}(\ell)$.

Table 2 summarizes the environment; Subsections III.B–III.G derive closed-form rules that reduce the equilibrium to a coupled spatial operator and a dynamic county marginal-q game.

B State and Beliefs

At the start of date t, the persistent spatial state consists of bounded, positive, measurable knowledge profiles $A_{At}(\ell)$ and $A_{Mt}(\ell)$ and installed route capacity $T_t(\ell)$, observed alongside aggregate sectoral productivity gaps $\mathbf{z}_t = (z_{At}, z_{Mt}, z_{Tt})'$ and the predetermined demographic, coalfield energy, manufacturing-wedge, and statutory project schedules in $\mathcal{E}_t \subset \mathcal{F}_t$. Only the active sector at ℓ conducts research there, whereas both sectors adopt discoveries from the spatial frontier; infrastructure built at date t enters operating capacity at $t+1$ and accelerates knowledge adoption from $t+1$ onward.

Aggregate productivity gaps follow a stationary first-order Markov process on a compact symmetric box $\mathcal{B}$:

$$\mathbf{z}_{t+1} \mid \mathbf{z}_t \sim \mathrm{TN}_{\mathcal{B}}(\boldsymbol{\rho}\mathbf{z}_t, \boldsymbol{\Sigma}), \tag{9}$$

where $\mathrm{TN}_{\mathcal{B}}$ is a multivariate Gaussian law truncated to $\mathcal{B}$ ($\pm c$ stationary untruncated standard deviations around zero, with $c = 3$ in the benchmark and $c = 4$ for robustness; Online Appendix S.C.1), $\boldsymbol{\rho} = \mathrm{diag}(\rho_A, \rho_M, \rho_T)$ with $|\rho_s| < 1$, and $\boldsymbol{\Sigma}$ is the positive definite innovation covariance matrix with $\sigma_s > 0$. The kernel has a unique stationary distribution π on $\mathcal{B}$, and effective productivity (18) is normalized by $\mathbb{E}_\pi[e^{z_i}]$ so that $e^{z_{it}}/\mathbb{E}_\pi[e^{z_i}]$ has unit stationary physical mean. Write $f(\mathbf{z}' \mid \mathbf{z})$ for this physical kernel and $M(\mathbf{m})$ for the conditional mean on $\mathcal{B}$ of a Gaussian centered at $\mathbf{m}$.

Belief formation follows the diagnostic expectations operator of Bordalo et al. (2018, 2020), which formalizes Kahneman and Tversky (1972)'s representativeness heuristic and underpins macroeconomic models of sentiment-driven cycles (Bianchi et al., 2024; Bordalo et al., 2026; L'Huillier et al., 2023; Maxted, 2023). Relative to the no-news reference forecast $\overline{\mathbf{z}}_t = \mathbb{E}_{t-1}[\mathbf{z}_t] = M(\boldsymbol{\rho}\mathbf{z}_{t-1})$, observed news $\boldsymbol{\varepsilon}_t = \mathbf{z}_t - \overline{\mathbf{z}}_t$ tilts perceived transition probabilities toward states whose likelihood increased most in light of $\boldsymbol{\varepsilon}_t$, weighted by the representativeness ratio $\Lambda_t(\mathbf{z}') = f(\mathbf{z}' \mid \mathbf{z}_t)/f(\mathbf{z}' \mid \overline{\mathbf{z}}_t)$:

$$\mathbb{E}_t^\theta[X] = \frac{\mathbb{E}_t[X\Lambda_t(\mathbf{z}_{t+1})^\theta]}{\mathbb{E}_t[\Lambda_t(\mathbf{z}_{t+1})^\theta]}, \qquad \theta \geq 0, \tag{10}$$

where $\theta = 0$ nests rational expectations and $\theta > 0$ indexes belief extrapolation. Because $\mathbf{z}_t$ is observed at the start of date t and local Bernoulli–Pareto discoveries are independent of $\mathbf{z}_{t+1}$, the

Table 2: Economic Environment

	Households
Lifetime utility	$W_t = \max_{\{\ell_s, c_A(s), c_M(s)\}_{s \geq t}} \mathbb{E}_t^\theta[\sum_{s=t}^{\infty} \beta^{s-t} U(c_A(s), c_M(s))]$
Period utility	$U(c_A, c_M) = (c_A - \gamma)^\eta c_M^{1-\eta}$
Budget	$p_A c_A + p_M c_M = w + b_t(\omega)$
Nonlabor income	$b_t = \bar{L}_t^{-1} \int_0^1 (R - p_M I + \Pi_F + \Pi_T + \Pi_\psi)\, d\ell$
	Production and Innovation
Bid rent	$R_i = \max_{\substack{0 \leq \phi_i < 1 \\ \bar{L}_i \geq 0}} \{\mathbb{E}_t^\theta[(p_i - r_i\tau_j)_+ Y_i] - (1+\psi_i) w L_i - \Psi_i(\phi_i; w, \ell)\}$
Production technology	$Y_i = e_i Z_i G_i L_i^{\alpha_i}$
Innovation expenditure	$\Psi_i(\phi_i; w, \ell) = w\left[f_i(\ell) + \frac{\chi_i}{1-\phi_i}\right]$ for $\phi_i > 0$, $\quad \Psi_i(0; w, \ell) = 0$
Research outcome	$\Pr(G_i > g \mid \phi_i) = \phi_i g^{-a_i}$, $\quad g \geq 1$
Knowledge after innovation	$A_{it}^+ = A_{it}[1 + e_{it}(G_{it} - 1)]$
	Transport Services
Operating profits	$\max_{L_T \geq 0} \mathbb{E}_t^\theta[\tau_j Y_T - w L_T]$
Transport technology	$Y_T = Z_T L_T^{\alpha_T} T^{1-\alpha_T}$
County capacity and demand	$K_j = \int_{\mathcal{C}_j} Y_T\, d\ell$, $\quad D_j = \int_{\mathcal{C}_j} \sum_i r_i (Y_i + \lvert H_i \rvert)\, d\ell$
	County Infrastructure Companies
Value function	$V_{jt}^\theta = \max_{0 \leq x_j \leq \bar{x}_{jt}} \{R_{jt} + \beta \mathbb{E}_t^\theta[V_{j,t+1}^\theta]\}$
Net rents	$R_{jt} = \mathbb{E}_t^\theta \int_{\mathcal{C}_j} [R_t - p_{Mt} I_t(\ell; \mathbf{x}_t)]\, d\ell$
Installed capacity	$T_{t+1} = T_t + \sum_j x_{jt} \iota_{jt}$
	State Transitions
Productivity	$Z_{it} = A_{it} e^{z_{it}} / \mathbb{E}_\pi[e^{z_i}]$, $\quad i \in \{A, M, T\}$
Sectoral shocks	$\mathbf{z}_{t+1} \mid \mathbf{z}_t \sim \mathrm{TN}_{\mathcal{B}}(\boldsymbol{\rho}\mathbf{z}_t, \boldsymbol{\Sigma})$
Available frontier	$F_{it}(\ell) = \sup_{r \in [0,1]} e^{-\delta \lvert \ell - r \rvert} A_{it}^+(r)$
Adoption	$A_{i,t+1} = A_{it}^+ + \left[1 - e^{-\lambda_0 (T_t / T_{t_0})^{1-\alpha_T}}\right](F_{it} - A_{it}^+)$
	Market Clearing
Residence	$\mathbb{E}_t^\theta[(w + b - \gamma p_A)/\mathcal{P}] = \bar{u}$
Goods	$\partial_\ell H_i = Y_i - n c_i - e_i \Psi_i / p_i - I_i - \kappa_0 e^{-T_t} \lvert H_i \rvert$
Labor and population	$n = \sum_i e_i L_i + L_T$, $\quad \int_0^1 n\, d\ell = \bar{L}_t$
Services	$D_j \leq K_j$, $\quad \tau_j \geq 0$, $\quad \tau_j (K_j - D_j) = 0$

Notes: Local and date arguments are suppressed; $i \in \{A, M\}$ unless stated otherwise, and j indexes the county. $e_i \in \{0, 1\}$ indicates the highest-bidding goods sector with land rent $R = \max_i R_i$; $f_A(\ell) = f_A$ and $f_M(\ell) = f_M + \xi(\ell)$; construction uses manufactured goods ($I_A = 0$, $I_M = I$). Borders are closed and the numeraire is $p_A(1, t) = 1$. Remaining definitions are in the text.

factor $\Lambda_t(\mathbf{z}_{t+1})^\theta$ cancels from (10) for any within-period payoff conditional on $(\Omega_t, \mathbf{x}_t)$. Exponentiating the Gaussian kernel yields the closed-form diagnostic one-step law and recursive forecast reference (Online Appendix S.C.1):

$$\mathbf{z}_{t+1} \overset{\theta}{\sim} \mathrm{TN}_{\mathcal{B}}(\mathbf{m}_t^\theta, \boldsymbol{\Sigma}), \qquad \mathbf{m}_t^\theta = \boldsymbol{\rho}\mathbf{z}_t + \theta\boldsymbol{\rho}(\mathbf{z}_t - \overline{\mathbf{z}}_t), \tag{11}$$

$$\overline{\mathbf{z}}_{t+1}^{\text{forecast}} = M(\mathbf{m}_t^\theta), \qquad \boldsymbol{\varepsilon}_{t+1}^{\text{forecast}} = \mathbf{z}_{t+1} - M(\mathbf{m}_t^\theta), \tag{12}$$

whereas in physical time at date $t+1$, the observed reference resets to $\overline{\mathbf{z}}_{t+1} = M(\boldsymbol{\rho}\mathbf{z}_t)$. Equation (11) delivers the two empirical regularities documented in Section II: forecasts overreact to current news with location shift $(1 + \theta)\boldsymbol{\rho}\boldsymbol{\varepsilon}_t$, and subsequent realizations systematically disappoint after positive

news ($\mathbf{m}_t^\theta - \boldsymbol{\rho}\mathbf{z}_t = \theta\boldsymbol{\rho}\boldsymbol{\varepsilon}_t$). The aggregate state is $\Omega_t = (A_{At}, A_{Mt}, T_t, \mathbf{z}_t, \bar{\mathbf{z}}_t, t)$ (Figure 5).

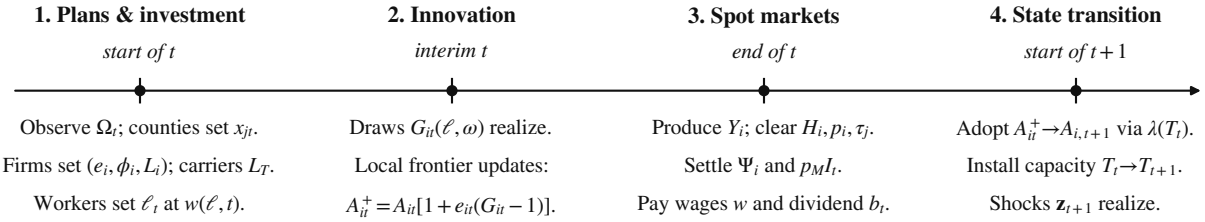


Figure 5: Within-Period Timing

Notes: County investment x_{jt}, producer plans (e_i, ϕ_i, L_i), carrier employment L_T, and worker residence ℓ_t at local contract wage $w(\ell, t)$ are committed before idiosyncratic discoveries $G_{it}(\ell, \omega)$ realize. Spot goods and service markets clear ex post, and completed capacity T_{t+1} and diffused knowledge $A_{i,t+1}$ enter at date $t+1$.

C Workers

Workers live where they work, supply one unit of labor inelastically, and hold equal shares in the national portfolio of land leases, goods producers, transport firms, and factor-wedge rebates, receiving common nonlabor income b_t in addition to the local contract wage $w(\ell, t)$. Goods are nonstorable, and households neither save nor borrow. Each period t, a worker commits residence $\ell_t \in [0, 1]$ before private discoveries at the fixed contract wage $w(\ell, t)$ and chooses spot consumption $(c_A(\ell, t), c_M(\ell, t))$ ex post to solve

$$\begin{aligned} W_t = \max_{\{\ell_s, c_A(s), c_M(s)\}_{s \geq t}} \; & \mathbb{E}_t^\theta \left[\sum_{s=t}^{\infty} \beta^{s-t} U(c_A(s), c_M(s)) \right] \\ \text{s.t. } & p_A(\ell_s, s) c_A(s) + p_M(\ell_s, s) c_M(s) = w(\ell_s, s) + b_s, \quad c_A(s) \geq \gamma, \; c_M(s) \geq 0, \end{aligned} \tag{13}$$

with nonhomothetic Stone–Geary period utility (Fajgelbaum and Redding, 2022; Herrendorf et al., 2014; Kongsamut et al., 2001):

$$U(c_A(\ell, t), c_M(\ell, t)) = (c_A(\ell, t) - \gamma)^\eta c_M(\ell, t)^{1-\eta}, \qquad \eta \in (0, 1), \quad \gamma > 0. \tag{14}$$

For $w + b_t \geq \gamma p_A$, spot optimization yields the Marshallian demands, marginal price index, and realized indirect utility:

$$\begin{aligned} c_A &= \gamma + \frac{\eta(w + b_t - \gamma p_A)}{p_A}, & c_M &= \frac{(1-\eta)(w + b_t - \gamma p_A)}{p_M}, \\ \mathcal{P}(p_A, p_M) &= \frac{p_A^\eta p_M^{1-\eta}}{\eta^\eta (1-\eta)^{1-\eta}}, & u_t(\ell) &= \frac{w(\ell, t) + b_t - \gamma p_A(\ell, t)}{\mathcal{P}(p_A(\ell, t), p_M(\ell, t))}. \end{aligned} \tag{15}$$

The subsistence floor $\gamma > 0$ generates Engel's law, shifting expenditure toward manufacturing as real income rises. The demands and $u_t(\ell)$ are defined where $w + b_t \geq \gamma p_A$; statewise feasibility at every site is part of the equilibrium definition (Definition 1) and is delivered by Condition 1 of Section III.G. Because residence is committed before private discoveries, workers evaluate locations by expected current indirect utility $\mathbb{E}_t^\theta[u_t(\ell)]$.

PROPOSITION 1: *If subsistence is feasible statewise at every site, expected indirect utility $\mathbb{E}_t^\theta[u_t(\ell)]$ is finite for every ℓ, and lifetime utility is finite, then, given the current price, wage and portfolio-income schedules, a worker's optimal dynamic residence choice maximizes expected current-period indirect utility.*[5] [Proof →]

Because annual relocation is costless and portfolio claims do not depend on residence, the continuation value of the worker's problem is invariant to current location, equalizing expected indirect utility across occupied sites:

$$\mathbb{E}_t^\theta[u_t(\ell)] = \bar{u}_t \qquad \text{where } n(\ell,t) > 0 \quad (\mathbb{E}_t^\theta[u_t(\ell)] \le \bar{u}_t \text{ elsewhere}), \tag{16}$$

$$w(\ell,t) = \frac{\bar{u}_t - \mathbb{E}_t^\theta\left[(b_t - \gamma p_A(\ell,t))/\mathcal{P}(p_A(\ell,t), p_M(\ell,t))\right]}{\mathbb{E}_t^\theta\left[1/\mathcal{P}(p_A(\ell,t), p_M(\ell,t))\right]}. \tag{17}$$

Because labor demand is strictly positive at every site for any finite contract wage (carriers hire by (49) whenever $\mathbb{E}_t[\tau_{j(\ell)}] > 0$, and otherwise the winning goods sector hires by (29) at a positive expected net price), every site is occupied in equilibrium, (16) holds with equality on all of $[0,1]$, and (17) determines the contract wage at every site.

D Production, Research, and Land

Competitive entrants in sector $i \in \{A, M\}$ bid for one-period pure site leases at $\ell \in [0,1]$.[6] Effective productivity combines permanent local knowledge $A_i(\ell,t)$ (or constant carrier frontier $A_T > 0$) with the aggregate productivity factor, and realized output Y_i uses committed labor $L_i(\ell,t)$ under diminishing returns:

$$Z_i(\ell,t) = A_i(\ell,t)e^{z_{it}}/\mathbb{E}_\pi[e^{z_i}], \quad i \in \{A,M\}, \qquad Z_T(t) = A_T e^{z_{Tt}}/\mathbb{E}_\pi[e^{z_T}], \tag{18}$$

$$Y_i(\ell,t,\omega) = e_i(\ell,t)Z_i(\ell,t)G_i(\ell,t,\omega)L_i(\ell,t)^{\alpha_i}, \qquad i \in \{A,M\}, \quad 0 < \alpha_A < \alpha_M < 1, \tag{19}$$

where $\alpha_A < \alpha_M$ reflects the higher land intensity of agriculture relative to manufacturing (Valentinyi and Herrendorf, 2008). Entrants announce (e_i, ϕ_i, L_i) taking state-contingent spot prices $p_i(\ell,t,\omega)$, county tolls $\tau_{j(\ell)}(t,\omega)$, and the local contract wage $w(\ell,t)$ as given, earning the net unit price $p_i(\ell,t,\omega) - r_i\tau_{j(\ell)}(t,\omega)$ ex post.

Manufacturing firms face a predetermined local operating-labor wedge,

$$\psi_A(\ell) = 0, \qquad \psi_M(\ell) = \bar{\psi}_M \omega_M(\ell), \qquad \bar{\psi}_M \ge 0, \quad \omega_M(\ell) > 0, \tag{20}$$

paying $(1 + \psi_M(\ell))w(\ell,t)$ per hired production worker while agriculture is undistorted ($\psi_A = 0$) and wedge receipts are rebated lump-sum to households. Holding $\omega_M(\ell)$ fixed makes transport improvements a source of first-order allocative efficiency gains as labor reallocates toward distorted industrial producers (Baqaee and Farhi, 2020; Hornbeck and Rotemberg, 2024).

Before private discoveries, an entrant in sector $i \in \{A,M\}$ chooses a research success probability $\phi_i \in [0,1)$. Building on the Pareto idea-flow representation of Kortum (1997) and Eaton and Kortum (2002), a successful draw yields a Pareto productivity multiplier with tail exponent $a_i > 1/(1-\alpha_i)$:

$$\Pr(G_i = 1 \mid \phi_i) = 1 - \phi_i, \tag{21}$$

$$\Pr(G_i > g \mid \phi_i) = \phi_i g^{-a_i}, \qquad g \ge 1, \quad a_i > 1. \tag{22}$$

[5] Proofs of the propositions are in Appendix A.

[6] Transport routes occupy narrow corridors and do not crowd out productive land.

Discoveries are spatially correlated. A success field $X^s_{it}(\ell)$ and a quality field $X^g_{it}(\ell)$ are stationary standard Gaussian Ornstein–Uhlenbeck processes on $[0,1]$, independent of each other, across sectors and dates, and of aggregate productivity gaps, with correlation length $d_0 > 0$; with Φ the standard normal distribution function, an entrant choosing ϕ at ℓ draws

$$\begin{aligned} G_i(\phi;\ell,\omega) &= 1 + \mathbf{1}_{\{\Phi(X^s_{it}(\ell))<\phi\}}\big[\Phi(X^g_{it}(\ell))^{-1/a_i} - 1\big], \\ \mathrm{Cov}(X^h_{it}(\ell), X^h_{it}(r)) &= e^{-|\ell-r|/d_0}, \qquad h \in \{s,g\}. \end{aligned} \tag{23}$$

Because $\Phi(X^h_{it}(\ell))$ is uniform at every site, (23) reproduces (21)–(22) at each ℓ, while neighboring sites draw correlated discoveries. All research intensities share one probability space: $G_i(\phi;\ell,\omega)$ is nondecreasing in ϕ, and $G_i(\phi_n;\ell,\omega) \to G_i(\phi;\ell,\omega)$ almost surely whenever $\phi_n \to \phi$.

Innovation requires a wage-indexed monetary expenditure committed before discovery and settled in the spot market by purchasing own-sector output locally:

$$\Psi_i(\phi_i; w, \ell) = \begin{cases} w\left[f_i(\ell) + \dfrac{\chi_i}{1-\phi_i}\right] & \text{if } \phi_i \in (0,1), \\ 0 & \text{if } \phi_i = 0, \end{cases} \tag{24}$$

$$q^R_i(\omega) = \frac{\Psi_i(\phi_i; w, \ell)}{p_i(\omega)}, \tag{25}$$

where $f_A(\ell) = f_A \geq 0$ and $f_M(\ell) = f_M + \xi(\ell) \geq 0$ combines fixed participation cost $f_i \geq 0$ with predetermined manufacturing energy overhead $\xi(\ell) = \omega_E e(\ell) \geq 0$ ($\omega_E \geq 0$ scaling the standardized coalfield energy exposure schedule $e(\ell)$ of Allen 2009), and $\chi_i > 0$ governs research convexity. Gross marketed output Y_i includes research goods, using local distribution services $r_i Y_i$ once, while only the knowledge transition carries discoveries to future periods.

Firm optimization and innovation threshold.— Risk-neutral firms measure payoffs in the numeraire $p_A(1,t) = 1$ and hold neither future land claims nor patents. Taking state-contingent goods prices $p_i(\omega)$, county tolls $\tau_{j(\ell)}(\omega)$, and the local contract wage w as given, a producer with hired labor L_i and discovery $G_i(\omega)$ chooses spot output utilization to maximize operating surplus:

$$\max_{0 \leq Y \leq Z_i G_i L_i^{\alpha_i}} \Big\{ (p_i - r_i \tau_{j(\ell)}) Y - (1+\psi_i) w L_i \Big\} = \big(p_i - r_i \tau_{j(\ell)}\big)_+ Z_i G_i L_i^{\alpha_i} - (1+\psi_i) w L_i, \tag{26}$$

where $[x]_+ = \max\{x, 0\}$, so full capacity $Y_i = e_i Z_i G_i L_i^{\alpha_i}$ is operated whenever $p_i(\omega) > r_i \tau_{j(\ell)}(\omega)$. Writing expected net unit revenue per efficiency unit as

$$m_i(\phi_i; \ell) = \mathbb{E}^\theta_t \Big[\big(p_i(\ell,\omega) - r_i \tau_{j(\ell)}(\omega)\big)_+ G_i(\phi_i; \ell, \omega) \Big], \tag{27}$$

the entrant maximizes expected land-bid surplus:

$$R_i(\ell) = \max_{\substack{0 \leq \phi_i < 1 \\ L_i \geq 0}} \Big\{ Z_i(\ell) m_i(\phi_i; \ell) L_i^{\alpha_i} - (1 + \psi_i(\ell)) w(\ell) L_i - \Psi_i(\phi_i; w(\ell), \ell) \Big\}. \tag{28}$$

First-order optimization in L_i yields the closed-form hiring rule and profiled scalar innovation problem:

$$L^*_i(\phi_i; w, \ell) = \left[\frac{\alpha_i Z_i(\ell) m_i(\phi_i; \ell)}{(1+\psi_i(\ell)) w(\ell)} \right]^{\frac{1}{1-\alpha_i}}, \tag{29}$$

$$R_i(w, \ell) = \max_{\phi_i \in [0,1)} \Big\{ \Lambda_i(\ell) w(\ell)^{-\frac{\alpha_i}{1-\alpha_i}} m_i(\phi_i; \ell)^{\frac{1}{1-\alpha_i}} - \Psi_i(\phi_i; w(\ell), \ell) \Big\}, \tag{30}$$

with $L_i^*(\phi_i; w, \ell) = 0$ if $m_i(\phi_i; \ell) = 0$ and $\Lambda_i(\ell) = (1-\alpha_i)Z_i(\ell)[\alpha_i Z_i(\ell)/(1+\psi_i(\ell))]^{\alpha_i/(1-\alpha_i)}$. On $\phi_i \in (0,1)$, Ψ_i is strictly convex with $\partial^2_{\phi\phi}\Psi_i = 2w(\ell)\chi_i/(1-\phi_i)^3 > 0$ and diverges as $\phi_i \uparrow 1$. By the envelope theorem, an interior research optimum $\tilde{\phi}_i(w,\ell) \in (0,1)$ satisfies

$$Z_i(\ell)L_i^*(\tilde{\phi}_i; w, \ell)^{\alpha_i}\partial_\phi m_i(\tilde{\phi}_i; \ell) = \frac{w(\ell)\chi_i}{\left(1-\tilde{\phi}_i(w,\ell)\right)^2}, \tag{31}$$

and for $\chi_i > \bar{\chi}_{it}$ the active objective is strictly concave on $(0,1)$ with at most one interior maximizer. Dividing (30) by $w(\ell)$ with baseline profitability index $\pi_i^0(w,\ell) \equiv \Lambda_i(\ell)[m_i(0;\ell)/w(\ell)]^{1/(1-\alpha_i)}$ and discovery gain $\Gamma_i(\phi_i;\ell) \equiv [m_i(\phi_i;\ell)/m_i(0;\ell)]^{1/(1-\alpha_i)} - 1$ gives

$$\frac{R_i(w,\ell)}{w(\ell)} = \pi_i^0(w,\ell) + \max\left\{0, \max_{\phi_i \in (0,1)}\left[\pi_i^0(w,\ell)\Gamma_i(\phi_i;\ell) - f_i(\ell) - \frac{\chi_i}{1-\phi_i}\right]\right\}. \tag{32}$$

Comparing $\tilde{\phi}_i(w,\ell)$ with zero research defines a unique reservation threshold $\underline{\pi}_i(\ell) \in (0,\infty]$ (Appendix A.6.1): innovation is active ($\phi_i^*(w,\ell) = \tilde{\phi}_i(w,\ell) > 0$) if and only if $\pi_i^0(w,\ell) > \underline{\pi}_i(\ell)$, and zero otherwise. Transport improvements $T_t(\ell)$ raise net unit prices $p_i(\ell,\omega) - r_i\tau_{j(\ell)}(\omega)$ while coalfield proximity lowers $\xi(\ell)$, pushing $\pi_M^0(w,\ell)$ above $\underline{\pi}_M(\ell)$ in industrializing counties. Since $\pi_i^0(w,\ell) \propto w(\ell)^{-1/(1-\alpha_i)}$, higher contract wages weakly lower $\phi_i^*(w,\ell)$ and strictly lower $L_i^*(\phi_i^*(w,\ell); w, \ell)$ (Appendix A.6.1).

Land competition and spatial sorting.— Competitive bidding assigns each site to the sector with the higher expected bid rent:

$$R(\ell) = \max\{R_A(\ell),\, R_M(\ell)\}, \qquad e_i(\ell) = \begin{cases} 1 & \text{if } R_i(\ell) = R(\ell), \\ 0 & \text{otherwise.} \end{cases} \tag{33}$$

Because $\alpha_A < \alpha_M$ and $\Lambda_i(\ell) \propto Z_i(\ell)^{1/(1-\alpha_i)}(1+\psi_i(\ell))^{-\alpha_i/(1-\alpha_i)}$, the sign of $R_A(w,\ell) - R_M(w,\ell)$ is governed by the elasticity- and wedge-adjusted comparative-advantage profile $\frac{1}{1-\alpha_M}\log Z_M(\ell) - \frac{1}{1-\alpha_A}\log Z_A(\ell) - \frac{\alpha_M}{1-\alpha_M}\log(1+\psi_M(\ell))$. Under Condition 2 of Section III.G, which requires this profile to rise steeply enough to dominate the spatial variation of wages, net prices, and research activity, $R_A(w,\ell) - R_M(w,\ell)$ changes sign exactly once, at an interior boundary $b_{ct} \in (0,1)$, assigning $[0, b_{ct}]$ to agriculture and $(b_{ct}, 1]$ to manufacturing; in the frictionless benchmark of Appendix A.6.3, where wages and prices are spatially uniform, an increasing profile is by itself sufficient for at most one crossing. At the winning plan, local employment, worker participation, output, and wedge transfers satisfy

$$n(\ell) = \sum_{i \in \{A,M\}} e_i(\ell)L_i(\ell) + L_T(\ell), \tag{34}$$

$$\bar{u}_t = \mathbb{E}_t^\theta\left[\frac{w(\ell) + b_t - \gamma p_A(\ell)}{\mathcal{P}(p_A(\ell), p_M(\ell))}\right], \tag{35}$$

and realized output, innovation expenditure, and wedge transfers satisfy

$$Y_i = e_i Z_i G_i L_i^{\alpha_i}, \qquad \Psi_i = \Psi_i(\phi_i^*(\ell); w(\ell), \ell), \qquad \Pi_\psi = \sum_i e_i \psi_i w L_i. \tag{36}$$

PROPOSITION 2: *Fix the inherited physical state, dated environment, and a common rule for selecting conditional private equilibria that depends only on the inherited state, the dated environment, the investment profile, and the date-t discoveries. Conditional on an investment profile, current household, firm, and carrier decisions, market prices, and county current dividends are invariant to θ. If two belief specifications select the same profile, the same private discoveries and sectoral shocks generate the same current allocation and next physical state.* [Proof →]

Because $\mathbb{E}_t^\theta$ in (10) reweights only future aggregate shocks $\mathbf{z}_{t+1}$ while leaving the conditional distribution of date-t private discoveries $G_{it}(\ell,\omega)$ unchanged, every within-period expectation conditional on $(\Omega_t, \mathbf{x}_t)$ reduces to rational expectations ($\mathbb{E}_t^\theta[\cdot] = \mathbb{E}_t[\cdot]$ in (17), (27), and (35)); diagnostic expectations affect the allocation solely through county infrastructure investment $\mathbf{x}_t^\theta$.[7]

After early-stage plans are committed, the active producer's realized discovery $G_{it}(\ell) \geq 1$ enters the public knowledge pool, updating local pre-diffusion technology (Cai et al., 2022):

$$A_{it}^+(\ell) = A_{it}(\ell)\big[1 + e_{it}(\ell)(G_{it}(\ell) - 1)\big]. \tag{37}$$

The inactive sector ($e_{it}(\ell) = 0$) retains $A_{it}(\ell)$ at this step, and transitory aggregate gaps $\mathbf{z}_t$ do not enter permanent knowledge. Because spillovers attenuate exponentially with distance (Berkes et al., 2025; Keller, 2002), the technology frontier available at ℓ is the pointwise spatial supremum

$$F_{it}(\ell) = \sup_{r\in[0,1]} e^{-\delta|\ell - r|} A_{it}^+(r), \qquad \delta > 0, \tag{38}$$

which is uniquely defined everywhere under pointwise entrant optimality and land sorting, and finite almost surely because the Gaussian fields generating the discoveries in (23) have continuous sample paths (Online Appendix Lemma S.C.1). Both sectors then close a fraction of their gap to $F_{it}(\ell)$ via the Nelson and Phelps (1966) catch-up law at a speed governed by installed route capacity (Berkes et al., 2026; Cinnirella et al., 2024):

$$A_{i,t+1}(\ell) = A_{it}^+(\ell) + (1 - e^{-\lambda_t(\ell)})[F_{it}(\ell) - A_{it}^+(\ell)], \tag{39}$$

$$\lambda_t(\ell) = \lambda_0 \left[\frac{T_t(\ell)}{T_{t_0}(\ell)}\right]^{1-\alpha_T}, \qquad \lambda_0 > 0, \tag{40}$$

where $T_{t_0}(\ell)$ is the initial reference network (half-life $\log(2)/\lambda_0$ years).

PROPOSITION 3: *For bounded positive post-discovery knowledge and finite nonnegative adoption rates, $A_{it}^+(\ell) \leq A_{i,t+1}(\ell) \leq F_{it}(\ell)$. Holding discoveries and reference levels fixed, more inherited capacity weakly raises next-period productivity. If no further discoveries occur and geography is fixed, the available frontier remains unchanged and*

$$F_{it}(\ell) - A_{i,t+h}(\ell) = [F_{it}(\ell) - A_{it}^+(\ell)] \exp\left[-\sum_{k=t}^{t+h-1} \lambda_k(\ell)\right], \qquad h \geq 1. \tag{41}$$

[Proof →]

Localized research in industrial hubs pushes out $F_{it}(\ell)$, while county transport investments raise $\lambda_t(\ell)$ and propagate discoveries across space. Because neither private entrants nor county promoters internalize how $T_t(\ell)$ accelerates future technology absorption, (39)–(40) generate a dynamic spatial adoption externality. Online Appendix S.C.2 proves that the frontier is finite with finite moments of every order below a_i and gives the exact conditional mean of next-period knowledge.

E Transport and Distribution

Moving goods over land requires both durable route infrastructure (Donaldson, 2018; Redding and Turner, 2015) and operating carriage services priced in local transport markets (Brancaccio et al.,

[7] Keeping within-period Bernoulli–Pareto discovery probabilities physical isolates the intertemporal network-investment channel, where construction must be committed a full period before operating returns realize.

2020). Transport firms combine installed route capacity $T(\ell,t)$ with hired labor $L_T(\ell,t)$ committed at the local contract wage $w(\ell,t)$ before private discoveries, selling operating services ex post at the county toll τ_j:

$$Y_T(\ell,t) = Z_T(t) L_T(\ell,t)^{\alpha_T} T(\ell,t)^{1-\alpha_T}, \qquad Z_T(t) > 0, \quad 0 < \alpha_T < 1. \tag{42}$$

Each county $\mathcal{C}_j$ operates a competitive operating-service market at a common toll $\tau_{j(\ell)}$: services produced anywhere in $\mathcal{C}_j$ meet distribution or transit demand anywhere within $\mathcal{C}_j$ without additional cost, whereas services cannot be dispatched across county boundaries.

Shipping good i across space incurs both physical transit degradation at rate $\kappa(\ell,t)$ and r_i units of county operating services per unit of distance at toll $\tau_{j(\ell)}(t)$, yielding the effective ad valorem transport-cost rate $\tilde{\kappa}_i(\ell,t)$ and spatial no-arbitrage law for signed trade flow $H_i(\ell,t)$ (positive rightward):

$$\tilde{\kappa}_i(\ell,t) = \kappa(\ell,t) + \frac{r_i \tau_{j(\ell)}(t)}{p_i(\ell,t)}, \tag{43}$$

$$|\partial_\ell p_i| \le \kappa p_i + r_i \tau_{j(\ell)}, \qquad \partial_\ell p_i = (\kappa p_i + r_i \tau_{j(\ell)}) \operatorname{sign}(H_i) \quad \text{if } H_i \neq 0. \tag{44}$$

Because (44) bounds the price gradient by the bounded quantity $\kappa p_i + r_i \tau_{j(\ell)}$, spot prices $p_i(\cdot,t)$ are Lipschitz, hence absolutely continuous, across space: a step in $\tau_{j(\ell)}$ across a county border changes the slope bound and kinks $\partial_\ell p_i$ without creating a price jump. Installed infrastructure lowers physical transit losses and bilateral iceberg delivery factors $d_t(\ell,r)$ exponentially:

$$\kappa(\ell,t) = \kappa_0 e^{-T(\ell,t)}, \qquad \kappa_0 > 0, \quad T(\ell,t) > 0, \tag{45}$$

$$d_t(\ell,r) = \exp\left\{ \kappa_0 \left| \int_r^\ell e^{-T_t(u)}\, du \right| \right\}. \tag{46}$$

Each unit of marketed output $Y_i(\ell,t)$ requires r_i units of county service for local origin distribution and r_i units per unit of distance shipped ($r_M = 1$, $r_A > 0$), so county-wide service demand and operating capacity are

$$D_j(\omega) = \int_{\mathcal{C}_j} \sum_{i \in \{A,M\}} r_i \left(Y_i(\ell,\omega) + |H_i(\ell,\omega)| \right) d\ell, \qquad K_j = \int_{\mathcal{C}_j} Y_T(\ell,t)\, d\ell. \tag{47}$$

Carrier optimization and county service clearing.— Before private discoveries, a competitive carrier at $\ell \in \mathcal{C}_j$ chooses $L_T(\ell)$ taking $T(\ell)$, $w(\ell)$, and the distribution of τ_j as given:

$$\max_{L_T \ge 0} \mathbb{E}_t^\theta \left[\tau_j Y_T - w L_T \right], \qquad Y_T = Z_T L_T^{\alpha_T} T^{1-\alpha_T}, \tag{48}$$

$$w = \alpha_T Z_T L_T^{\alpha_T - 1} T^{1-\alpha_T}\, \mathbb{E}_t^\theta[\tau_j] \quad \Longleftrightarrow \quad L_T^*(w,\ell) = \left[\frac{\alpha_T Z_T\, \mathbb{E}_t^\theta[\tau_{j(\ell)}]}{w(\ell)} \right]^{\frac{1}{1-\alpha_T}} T(\ell), \tag{49}$$

where $\mathbb{E}_t^\theta[\tau_j] = \mathbb{E}_t[\tau_j]$ conditional on $(\Omega_t, \mathbf{x}_t)$ by Proposition 2. After private discoveries realize, predetermined county capacity K_j clears against county service demand D_j via complementary slackness, and carrier operating surplus accrues to the household portfolio:

$$D_j \le K_j, \qquad \tau_j \ge 0, \qquad \tau_j (K_j - D_j) = 0, \tag{50}$$

$$\Pi_T(\ell,t) = \tau_{j(\ell)}(t) Y_T(\ell,t) - w(\ell,t) L_T(\ell,t). \tag{51}$$

F County Infrastructure Companies

Unlike centralized national network planners (Allen and Arkolakis, 2022; Fajgelbaum and Schaal, 2020), historical British turnpikes, canals, and railways were promoted by local county companies through parliamentary acts (Bogart, 2014). Each county $j \in \{1, \dots, J\}$ has an infrastructure company holding a long-lived claim on land-lease rents inside $\mathcal{C}_j$. Roads, waterways, and railways accumulate into a composite route stock $T(\ell, t)$ under one-period time-to-build (Kydland and Prescott, 1982): before private discoveries at date t, company j commits completion intensity $x_{jt} \in [0, \bar{x}_{jt}]$ within the resource-feasible statutory box $\mathcal{X}_t = \prod_j [0, \bar{x}_{jt}] \subseteq \prod_j [0, o_{jt}]$ ($o_{jt} \in \{0, 1\}$), and completed increments enter operating route capacity at date $t+1$:

$$x_{jt} \in [0, \bar{x}_{jt}], \qquad \operatorname{supp} \iota_{jt} \subseteq \mathcal{C}_j, \tag{52}$$

$$T_{t+1}(\ell) = T_t(\ell) + \sum_{j=1}^{J} x_{jt} \iota_{jt}(\ell), \qquad T_0(\ell) \geq \underline{T} > 0. \tag{53}$$

Construction uses manufactured goods with quadratic adjustment costs on the county engineering profile (e_{jt}, k_{jt}) ($\operatorname{supp} k_{jt} \subseteq \mathcal{C}_j$, $e_{jt} > 0$):

$$I_t(\ell; \mathbf{x}_t) = \sum_{j=1}^{J} m_I e_{jt} k_{jt}(\ell) \left(x_{jt} + \frac{\chi_I}{2} x_{jt}^2 \right), \qquad m_I > 0, \quad \chi_I > 0, \tag{54}$$

purchased at the realized spot price $p_M(\ell, t, \omega)$. Full-package physical increments grow at two percent per year, $\iota_{jt} = 1.02^{t-t_0} \iota_{j,t_0}$; after the last observed schedule year $t = 1881$, each county repeats its 1881 opportunity $o_{j,1881}$ and engineering profile $(e_j k_j)_{1881}$ at the same (m_I, χ_I) while physical increments continue to grow as $\iota_{j,1881+h} = 1.02^h \iota_{j,1881}$.

Promoter optimization, marginal-q, and stage Nash equilibrium.— Taking rival counties' intensities $\mathbf{x}_{-j,t}$ as given, company j's current payoff is expected county land rent net of spot manufacturing construction expenditure:

$$R_{jt}(\mathbf{x}; \Omega_t) = \mathbb{E}_t^\theta \left[\int_{\mathcal{C}_j} \{ R_t(\ell; \mathbf{x}, \Omega_t) - p_{Mt}(\ell, \omega; \mathbf{x}, \Omega_t) I_t(\ell; \mathbf{x}) \} \, d\ell \right]. \tag{55}$$

By Proposition 2, current returns conditional on $(\Omega_t, \mathbf{x})$ depend only on physical private discoveries ($\mathbb{E}_t^\theta = \mathbb{E}_t$ in (55)), whereas future aggregate shocks enter the continuation value under $\mathbb{E}_t^\theta$ with perceived next-period reference state $\overline{\mathbf{z}}_{t+1}^{\text{forecast}} = M(\mathbf{m}_t^\theta)$ from (12) (while the realized physical state at date $t+1$ resets to $\overline{\mathbf{z}}_{t+1} = M(\boldsymbol{\rho} \mathbf{z}_t)$):

$$\Pi_{jt}^\theta(\mathbf{x}; \Omega_t) = R_{jt}(\mathbf{x}; \Omega_t) + \beta \mathbb{E}_t^\theta \left[V_{j,t+1}^\theta(\Omega_{t+1}) \right], \qquad V_{jt}^\theta(\Omega_t) = \Pi_{jt}^\theta \left(\mathbf{x}_t^\theta(\Omega_t); \Omega_t \right), \tag{56}$$

$$x_{jt}^\theta(\Omega_t) \in \arg \max_{0 \leq u \leq \bar{x}_{jt}} \Pi_{jt}^\theta(u, \mathbf{x}_{-j,t}^\theta(\Omega_t); \Omega_t), \qquad j = 1, \dots, J. \tag{57}$$

If the pure-strategy Nash set $\mathcal{N}_t^\theta$ contains multiple profiles, parliamentary coordination selects the profile maximizing $\sum_{j=1}^J \Pi_{jt}^\theta$ (with deterministic tie-breaking). Because the aggregate spatial knowledge frontier and future knowledge trajectories $(A_{As}, A_{Ms})_{s \geq t+1}$ reflect atomistic firm discoveries and frontier adoption across the national continuum, each county company takes the aggregate knowledge path as external to its own corridor completion choice x_{jt} when evaluating continuation values. Where differentiable, county j's perceived marginal continuation value q_{jt}^θ (differentiating

with respect to own route capacity T_{t+1} at the given knowledge path) and marginal construction cost c_{jt} satisfy (Hayashi, 1982):

$$q^{\theta}_{jt}(\mathbf{x}) = \beta \partial_{x_j} \mathbb{E}^{\theta}_t[V^{\theta}_{j,t+1}(\Omega_{t+1})], \qquad c_{jt}(\mathbf{x}) = -\partial_{x_j} R_{jt}(\mathbf{x}; \Omega_t), \tag{58}$$

$$q^{\theta}_{jt}(\mathbf{x}^{\theta}_t) = c_{jt}(\mathbf{x}^{\theta}_t) \tag{59}$$

at an interior best response $x^{\theta}_{jt} \in (0, \bar{x}_{jt})$, with complementary slackness at $\{0, \bar{x}_{jt}\}$. Because every engineering profile lies inside its own county ($\operatorname{supp} k_{kt} \subseteq \mathcal{C}_k$), county j pays only for its own construction, and (55) splits into gross county land rent and a construction bill, $R_{jt}(\mathbf{x}; \Omega_t) = \widetilde{R}_{jt}(\mathbf{x}) - m_I(x_{jt} + \frac{\chi_I}{2} x^2_{jt}) D^I_{jt}(\mathbf{x})$, with

$$\widetilde{R}_{jt}(\mathbf{x}) = \int_{\mathcal{C}_j} R_t(\ell; \mathbf{x}, \Omega_t)\, d\ell, \qquad D^I_{jt}(\mathbf{x}) = \int_{\mathcal{C}_j} \mathbb{E}_t[p_{Mt}(\ell, \omega; \mathbf{x}, \Omega_t)]\, e_{jt} k_{jt}(\ell)\, d\ell > 0, \tag{60}$$

where D^I_{jt} is the county construction-price index. Differentiating gives the marginal construction cost c_{jt} and, collecting the marginal benefits of capacity, the gross marginal value $\widetilde{q}^{\theta}_{jt}$:

$$c_{jt} = m_I(1 + \chi_I x_{jt}) D^I_{jt} - \partial_{x_j} \widetilde{R}_{jt} + m_I\Big(x_{jt} + \frac{\chi_I}{2} x^2_{jt}\Big) \partial_{x_j} D^I_{jt},$$

$$\widetilde{q}^{\theta}_{jt} = q^{\theta}_{jt} + \partial_{x_j} \widetilde{R}_{jt} - m_I\Big(x_{jt} + \frac{\chi_I}{2} x^2_{jt}\Big) \partial_{x_j} D^I_{jt}.$$

Condition (59) therefore reads $\widetilde{q}^{\theta}_{jt} = m_I(1+\chi_I x_{jt}) D^I_{jt}$, and with complementary slackness it becomes the clipped investment rule

$$x^{\theta}_{jt} = \min\left\{\bar{x}_{jt},\ \max\left\{0,\ \frac{1}{\chi_I}\left[\frac{\widetilde{q}^{\theta}_{jt}(\mathbf{x}^{\theta}_t)}{m_I D^I_{jt}(\mathbf{x}^{\theta}_t)} - 1\right]\right\}\right\}. \tag{61}$$

A county builds only when the marginal value of capacity per unit of construction outlay exceeds one, and lower adjustment convexity χ_I makes buildout more responsive to that ratio. Diagonal dominance of own adjustment-cost curvature over cross-county general-equilibrium spillovers yields a unique stage equilibrium (Vives, 1999):

PROPOSITION 4: *Fix $(\Omega_t, \mathcal{E}_t)$ and $\theta \geq 0$ on a nonempty compact box $\mathcal{X}_t = \prod_j [0, \bar{x}_{jt}]$ where every profile has a unique admissible private allocation and continuous county payoffs Π^{θ}_{jt}. If each Π^{θ}_{jt} is concave in x_{jt}, $\mathcal{N}^{\theta}_t$ is nonempty and compact. If, in addition, Π^{θ}_{jt} is C^2 and there exist $m_j > 0$, $L_{jk} \geq 0$, and $v_j > 0$ on $\mathcal{X}_t$ such that*

$$-\partial_{jj}\Pi^{\theta}_j \geq m_j > 0, \qquad |\partial_{jk}\Pi^{\theta}_j| \leq L_{jk} \quad (k \neq j), \tag{62}$$

$$\gamma_I = \max_j \frac{\sum_{k \neq j} L_{jk} v_k}{m_j v_j} < 1, \tag{63}$$

then the county Nash equilibrium is unique and simultaneous best responses converge to it from every initial profile in $\mathcal{X}_t$. If instead Π^{θ}_{jt} is C^1 and the pseudogradient $\mathbf{g}^{\theta}_t = (\partial_{x_j}\Pi^{\theta}_{jt})_j$ is strongly monotone on $\mathcal{X}_t$, $\sum_j (x_j - y_j)[g^{\theta}_{jt}(\mathbf{y}) - g^{\theta}_{jt}(\mathbf{x})] \geq \mu \|\mathbf{x} - \mathbf{y}\|^2$ for some $\mu > 0$, the Nash equilibrium is likewise unique. In either case, if Π^{θ}_{jt} is jointly measurable in Ω_t, the selection $\Omega_t \mapsto \mathbf{x}^{\theta}_t(\Omega_t)$ is measurable. [Proof →]

By Proposition 2, θ enters the dated allocation solely through $q^{\theta}_{jt}(\mathbf{x})$ in (59). Differentiating the diagnostic kernel (10)–(11) holding $V^{\theta}_{j,t+1}$ fixed shows that current news $\boldsymbol{\varepsilon}_t = \mathbf{z}_t - \bar{\mathbf{z}}_t$ shifts

perceived marginal-q directly and—together with $dq^\theta_{jt}/d\theta = \partial_\theta q^\theta_{jt}\big|_{V_{t+1}} + \beta \mathbb{E}^\theta_t[\partial_{x_j}\partial_\theta V^\theta_{j,t+1}]$—shifts interior buildout ($\mathcal{I}_t = \{j : x^\theta_{jt} \in (0, \bar{x}_{jt})\}$) according to

$$\begin{aligned} \partial_\theta q^\theta_{jt}(\mathbf{x})\Big|_{V_{t+1}} &= \beta \operatorname{Cov}^\theta_t\Big(\partial_{x_j} V^\theta_{j,t+1}(\Omega_{t+1}),\, \mathbf{z}'_{t+1}\boldsymbol{\Sigma}^{-1}\boldsymbol{\rho}\boldsymbol{\varepsilon}_t\Big), \\ \frac{d\mathbf{x}^\theta_{t,\mathcal{I}_t}}{d\theta} &= -\big[D_{\mathbf{x}_{\mathcal{I}_t}}(\mathbf{q}^\theta_{t,\mathcal{I}_t} - \mathbf{c}_{t,\mathcal{I}_t})\big]^{-1} \frac{d\mathbf{q}^\theta_{t,\mathcal{I}_t}(\mathbf{x}^\theta_t)}{d\theta}. \end{aligned} \tag{64}$$

Because each county company captures only local land rents inside $\mathcal{C}_j$ and takes aggregate knowledge evolution as external, its private marginal-q omits three positive social returns to route capacity: (i) cross-county market-access and transit savings $\sum_{k\neq j} \partial_{x_j}\Pi^0_{kt} > 0$ (Allen and Arkolakis, 2022; Donaldson and Hornbeck, 2016; Fajgelbaum and Schaal, 2020; Fogel, 1964; Krugman, 1991); (ii) the allocative-efficiency gain $\int_0^1 \psi_M(\ell) w(\ell) \partial_{x_j} L_{Mt}(\ell)\, d\ell > 0$ from reallocating labor into wedge-distorted manufacturing (Baqaee and Farhi, 2020; Hornbeck and Rotemberg, 2024; Hsieh and Klenow, 2009; Restuccia and Rogerson, 2008); and (iii) faster spatial knowledge diffusion (Aghion and Howitt, 1992; Desmet and Rossi-Hansberg, 2014; Lucas, 1988; Mokyr, 2005; Nelson and Phelps, 1966; Romer, 1990), since higher $T_s(\ell)$ at $s \geq t+1$ raises next-period knowledge at given discoveries by

$$\frac{\partial A_{i,s+1}(\ell)}{\partial T_s(\ell)} = (1-\alpha_T)\frac{\lambda_s(\ell)}{T_s(\ell)} e^{-\lambda_s(\ell)}\big[F_{is}(\ell) - A^+_{is}(\ell)\big] \geq 0, \tag{65}$$

which is strictly positive wherever the available frontier exceeds local technology. Rational promoters ($\theta = 0$) therefore omit three positive social returns and, to that extent, underinvest relative to a planner who internalizes them, whereas favorable manufacturing and transport news that raises the perceived continuation value, $q^\theta_{jt} > q^0_{jt}$ in (64), expands irreversible network buildout under diagnostic expectations ($\theta > 0$).

G Equilibrium and State Transitions

Fix $(\Omega_t, \mathcal{E}_t)$ and a feasible county investment profile $\mathbf{x}_t \in \mathcal{X}_t$. Before private discoveries, pre-discovery plans $(e_i, \phi_i, L_i, L_T, n, w)$ clear local and national labor markets, while ex-post goods flows $H_i(\ell, t)$ satisfy spatial conservation with closed borders:

$$n = \sum_{i\in\{A,M\}} e_i L_i + L_T, \tag{66}$$

$$\int_0^1 n\, d\ell = \bar{L}_t, \tag{67}$$

$$\partial_\ell H_i = Y_i - n c_i - e_i \frac{\Psi_i}{p_i} - I_i - \kappa|H_i|, \qquad H_i(0,t) = H_i(1,t) = 0, \quad i \in \{A, M\}. \tag{68}$$

Realized producer net profit Π_F, contractual land rent R, origin carrier surplus Π_T, manufacturing wedge rebates Π_ψ, net land rent R^{net}, and the common portfolio dividend b_t are

$$\Pi_F = \sum_i e_i \Big\{(p_i - r_i\tau_{j(\ell)})Y_i - (1+\psi_i) w L_i - \Psi_i(\phi_i; w, \ell)\Big\} - R, \tag{69}$$

$$R = \sum_i e_i R_i, \qquad \Pi_T = \tau_{j(\ell)} Y_T - w L_T, \qquad \Pi_\psi = \sum_i e_i \psi_i w L_i, \tag{70}$$

$$R^{\text{net}} = R - p_M I, \tag{71}$$

$$b_t = \frac{1}{\bar{L}_t}\int_0^1 \{R^{\text{net}} + \Pi_F + \Pi_T + \Pi_\psi\}\, d\ell. \tag{72}$$

Free entry implies $\mathbb{E}_t[\Pi_F] = 0$, while ex-post windfalls $\Pi_F(\ell, \omega)$ accrue to households. Carrier tolls and wedge rebates are pecuniary transfers, whereas research goods Ψ_i/p_i, construction I, and iceberg degradation $\kappa|H_i|$ absorb physical output.

DEFINITION 1 (Conditional dated equilibrium): Given $(\Omega_t, \mathcal{E}_t, \mathbf{x}_t)$, a conditional dated equilibrium consists of pre-discovery plans $(e_i, \phi_i, L_i, L_T, n, w, R, I)$ and a measurable post-discovery family $(Y_i, c_i, p_i, \tau_j, H_i, b_t)$ such that: (i) entrants solve (28), land is assigned by (33), and utilization solves (26); (ii) households choose residence by (16) and consumption by (15); (iii) carriers solve (48) and county services clear by (50); and (iv) for almost every realization, spatial no-arbitrage, goods and labor clearing, and (72) hold. At absolutely continuous versions of $p_i(\cdot, \omega)$ and $H_i(\cdot, \omega)$ (Lipschitz, by (44)) and continuous $w(\cdot)$ on $[0, 1]$ (with $p_A(1, t) = 1$), entrant optimality (28) and pure-site sorting (33) hold at every $\ell \in [0, 1]$ (breaking isolated ties to the left sector), differential flow and no-arbitrage relations hold almost everywhere, subsistence is feasible statewise, and expectations in agent objectives are finite.

In a competitive continuum economy (Aumann, 1966), individual entrants and carriers evaluate alternative plans at the quoted price and contract-wage functions (w, p, τ) without recomputing local market clearing, while equilibrium announcements jointly clear labor, goods, and county service markets.

PROPOSITION 5: *For every realization with absolutely continuous prices and flows satisfying local labor clearing, spatial goods balance, spatial no-arbitrage, county service clearing* (50)*, and closed national boundaries, the ledgers* (69)–(72) *imply the aggregate household budget identity. By Walras' Law, there is no additional independent national resource constraint.* [Proof →]

Building on classical general equilibrium analysis (Arrow and Hahn, 1971) and the nonlinear spatial operator representation of Allen and Arkolakis (2014) and Allen et al. (2024), substituting the closed-form household demand and mobility rules (15)–(17), producer hiring and research rules (29)–(31), land sorting (33), and carrier hiring (49) into market clearing (66)–(72) reduces the conditional dated equilibrium to a fixed point $(w, p, \tau) = \mathcal{T}_t(w, p, \tau)$ on the quote domain $\mathcal{W}_t \times \mathcal{P}_t \times \mathcal{M}_t$ of positive continuous wages $w \in C([0, 1]; [\underline{w}_t, \overline{W}_t])$, statewise continuous spot prices $p = (p_A, p_M)$ $(p_A(1, \omega) = 1)$, and nonnegative county tolls $\tau = (\tau_1, \ldots, \tau_J)$ with positive expected tolls $\mathbb{E}_t[\tau_j] \in [\underline{\tau}_{jt}, \overline{\tau}_{jt}]$ $(\underline{\tau}_{jt} > 0)$:

$$\begin{aligned} \widetilde{w}(\ell) = \big(\mathcal{T}_t^w(w, p, \tau)\big)(\ell) &= \frac{\bar{u}_t(w, p, \tau) - \mathbb{E}_t\big[(b_t(\omega; w, p, \tau) - \gamma p_A(\ell, \omega))/\mathcal{P}(\ell, \omega)\big]}{\mathbb{E}_t\big[1/\mathcal{P}(\ell, \omega)\big]}, \\ \big(\widetilde{p}(\cdot, \omega), \widetilde{\tau}(\omega)\big) &= \mathcal{T}_t^p(w, p, \tau; \omega), \\ \bar{L}_t &= \int_0^1 \Big[\sum_{i \in \{A, M\}} e_i(\ell) L_i^*\big(\phi_i^*(\widetilde{w}(\ell), \ell); \widetilde{w}(\ell), \ell\big) + L_T^*(\widetilde{w}(\ell), \ell)\Big] d\ell, \end{aligned} \tag{73}$$

where national labor clearing at $\widetilde{w} = \mathcal{T}_t^w(w, p, \tau)$ pins down $\bar{u}_t(w, p, \tau)$ and $\mathcal{T}_t^p(w, p, \tau; \omega)$ maps predetermined capacities and construction demand $I_t(\cdot; \mathbf{x}_t)$ into the unique spot prices and county tolls supporting the strictly concave Negishi-weighted log-surplus spatial dispatch problem under (44), (50), and (68)–(72).

Existence and uniqueness of the conditional dated equilibrium rest on three primitive conditions:

1. *(Resource feasibility and finite innovation tails.)* On the contract-wage collar $[\underline{w}_t, \overline{W}_t]$ where zero-research labor demand crosses $\bar{L}_t$, net deliverable agricultural capacity exceeds subsistence $(\underline{Y}_{At} > \gamma \bar{L}_t)$, net deliverable manufacturing capacity exceeds construction demand $(\underline{Y}_{Mt} > \int_0^1 I_t(\ell; \mathbf{x}_t)\, d\ell)$, county carrier capacity covers distribution demand uniformly on $\mathcal{X}_t$, and $a_i > 1/(1 - \alpha_i)$ for $i \in \{A, M\}$.

2. *(Innovation-cost convexity and Ricardian single-crossing.)* $\chi_i > \bar{\chi}_{it}$ (Appendix A.6.1) so active research (30) is strictly concave on $(0,1)$, $\pi_i^0(w(\ell),\ell) - \underline{\pi}_i(\ell)$ crosses zero transversally at most finitely many times when $f_i > 0$, and $\partial_\ell[\frac{1}{1-\alpha_M}\log Z_{Mt}(\ell) - \frac{1}{1-\alpha_A}\log Z_{At}(\ell) - \frac{\alpha_M}{1-\alpha_M}\log(1+\psi_M(\ell))] > 0$ dominates spatial factor-price and toll gradients on $\mathcal{U}_t$ so the bid-rent difference $R_A(\ell) - R_M(\ell)$ satisfies $R_A(0) > R_M(0)$, $R_A(1) < R_M(1)$, and $\partial_\ell[R_A(b_{ct}) - R_M(b_{ct})] < 0$ at a unique cutoff $b_{ct} \in (0,1)$.

3. *(Norm contraction.)* In a weighted Banach norm $\|\cdot\|_{\mathcal{U}_t}$, the Jacobian $\mathcal{D}_t(u)$ of $\mathcal{T}_t$ satisfies $\sup_{u\in\mathcal{U}_t}\|\mathcal{D}_t(u)\|_{\mathcal{U}_t} \le \bar{\rho}_t < 1$, so direct local labor-demand elasticities $1/(1-\alpha_i)$ dominate indirect spatial trade, toll, and dividend feedbacks.

PROPOSITION 6: *Consider the economy of Definition 1 with $0 < \alpha_A < \alpha_M < 1$, $\alpha_T \in (0,1)$, bounded positive piecewise-continuous (A_{At}, A_{Mt}), $\eta \in (0,1)$, $\gamma > 0$, and $\phi_{it}(\ell) \in [0,1)$.*

1. (Existence, active-research threshold, and pure-site sorting.) *Under Conditions 1 and 2, every $\mathbf{x}_t \in \mathcal{X}_t$ admits a regular conditional dated equilibrium in $\mathcal{U}_t$. At the continuous price and wage versions, optimal research $\phi_{it}^*(\ell) \in [0,1)$ is uniquely determined at every $\ell \in [0,1]$ by the threshold rule (31)–(32), and land partitions at a unique cutoff $b_{ct} \in (0,1)$ with agriculture on $[0, b_{ct}]$ and manufacturing on $(b_{ct}, 1]$, uniquely determining $A_{it}^+(\ell,\omega)$ and the almost surely finite frontier $F_{it}(\ell,\omega)$ (Online Appendix Lemma S.C.1) everywhere on $[0,1]$.*

2. (Global uniqueness.) *Under Condition 3, the regular conditional dated equilibrium is globally unique for each $\mathbf{x}_t \in \mathcal{X}_t$.*

3. (Constructive verification from primitives.) *Around the closed-form frictionless spatial benchmark, backward county food-anchor induction and regular perturbation verify a stock-uniform wage collar $[\underline{w}_{\text{nat}}, \overline{W}_{\text{nat}}]$ and Conditions 1–3 directly from primitives across all admissible county infrastructure paths $T_t(\ell) = T_0(\ell) + s_{j(\ell),t}(\ell) \ge T_0(\ell)$ (Appendix A.6.3).* [Proof →]

DEFINITION 2 (Dynamic equilibrium): Given Ω_0, diagnosticity $\theta \ge 0$, $\{\mathcal{E}_t\}_{t\ge 0}$, and the physical shock laws, a dynamic equilibrium is an adapted process of states, investments, conditional dated equilibria, and county values such that: (i) each dated allocation satisfies Definition 1; (ii) county policies satisfy (57) under values (56) and perceived forecast-reference update (12), taking the aggregate knowledge trajectory (A_{At}, A_{Mt}) as given; and (iii) realized states follow (9), (37)–(39), and (53) with observed reference $\bar{\mathbf{z}}_{t+1} = M(\boldsymbol{\rho}\mathbf{z}_t)$.

Substituting the unique conditional dated equilibrium rent and manufacturing price schedules from Proposition 6 into (55) yields county j's reduced stage payoff

$$R_{jt}(\mathbf{x}_t;\Omega_t) = \widetilde{R}_{jt}(\mathbf{x}_t) - m_I\left(x_{jt} + \frac{\chi_I}{2}x_{jt}^2\right)D_{jt}^I(\mathbf{x}_t),$$

with gross county land rent $\widetilde{R}_{jt}$ and construction-price index D_{jt}^I from (60), county mean route stock $\overline{T}_{j,t+1} = \overline{T}_{jt} + d_{jt}x_{jt}$ ($\overline{T}_{jt} = |\mathcal{C}_j|^{-1}\int_{\mathcal{C}_j} T_t(\ell)\,d\ell$ and $d_{jt} = |\mathcal{C}_j|^{-1}\int_{\mathcal{C}_j}\iota_{jt}(\ell)\,d\ell$ by (53)) and knowledge transition (37)–(39). After 1881, household mass remains at $\bar{L}_{1881+h} = \bar{L}_{1881}$ while the 1881 project schedule repeats with physical package growth $g = 1.02$. Write $\bar{Z}_T \equiv A_T/\mathbb{E}_\pi[e^{z_T}]$, $\varepsilon = e^{-z_{Tt}/\alpha_T}/(\bar{Z}_T)^{1/\alpha_T}$ for the carrier-friction scale, and $D_{\min} > 0$ for the lower bound on the construction-price index D_{jt}^I.

PROPOSITION 7: *Suppose the conditions of Proposition 6 hold along feasible state histories.*

1. (Finite- and infinite-horizon Markov perfect equilibrium.) *Whenever own adjustment-cost convexity $m_I \chi_I D_{\min}$ dominates cross-county price and continuation spillovers, the J-county stage pseudogradient is strongly monotone on $\mathcal{X}_t$ with modulus $\mu = \frac{1}{8} m_I \chi_I D_{\min} > 0$, attenuating cross-county policy sensitivity to inherited county stock $\overline{T}_{it}$:*

$$\left\| \frac{\partial \mathbf{x}_t^{(H),\theta}}{\partial \overline{T}_{it}} \right\| \leq \frac{C_u \varepsilon}{\mu \min_r \overline{T}_{rt}^{(1+\alpha_T)/\alpha_T}}. \tag{74}$$

For every finite horizon $H \geq 1$ and $\theta \in [0, \bar{\theta}]$, backward induction yields a unique pure-strategy Markov perfect equilibrium $(\mathbf{x}_t^{(H),\theta}, \mathbf{V}_t^{(H),\theta})_{t=0}^{H}$, which converges geometrically as $H \to \infty$ to a unique infinite-horizon dynamic equilibrium $(\mathbf{x}_t^{\theta}, \mathbf{V}_t^{\theta})_{t \geq 0}$ whenever the state-weight drift κ and policy sensitivity satisfy $\beta\kappa + BL_a < 1$ (Theorem A.1).

2. (Constructive verification on the rescaled corridor.) *Under the terminal schedule with $\beta g^{1-\alpha_T} < 1$, regular perturbation around the frictionless benchmark verifies stage-game strong monotonicity, horizon-invariant continuation concavity, and geometric convergence to a unique interior infinite-horizon equilibrium on a forward-invariant rescaled stock corridor (Appendix A.7).* [Proof →]

IV Quantitative Analysis

This section estimates the model on historical evidence from England and Wales over 1710–1881 and uses the estimated model to quantify the contribution of diagnostic expectations to transport accumulation, structural transformation, real wages, prices, and frontier knowledge during the industrial takeoff (Broadberry et al., 2014; Crafts and Harley, 1992). Spatial allocations are computed on $N = 500$ locations partitioned into $J = 10$ county jurisdictions, solving at each annual date for the conditional dated equilibrium and the Markov perfect county investment profile.

A From Historical Districts to Model Space

Because England and Wales is predominantly north–south oriented, latitude provides a natural one-dimensional ordering of regional economic activity on $\ell \in [0, 1]$ (Bogart et al., 2022; Trew, 2020). For each of twenty-five equal-latitude bands, historical employment and route quantities are summed across the band's east–west extent and divided by land width, with $\ell = 0$ at the southern coast and $\ell = 1$ at the northern border.

This projection captures the relative positions of metropolitan London in the South, the West Midlands, and the northern coalfield belt, together with the reallocation of labor across them, while abstracting from east–west variation within a latitude band (Online Appendix S.B).

B Data and External Calibration

Aggregate household mass $\bar{L}_t$ follows the annual England and Wales population series of Thomas and Dimsdale (2017) over 1710–1881, normalized to $\bar{L}_{1710} = 100$ and held constant after 1881 ($\bar{L}_{1881+h} = \bar{L}_{1881}$). Installed transport capacity is reconstructed annually across the twenty-five latitude bands from turnpike, waterway, and railway GIS records. Letting M_{hy}^r, M_{hy}^w, and M_{hy}^g denote permanent route miles of turnpike road (Rosevear et al., 2026), navigable waterway (Cambridge Group for the History of Population and Social Structure, 2016a), and railway (Cambridge Group for the History of Population and Social Structure, 2016b; Coll Ardanuy et al., 2021; Mitchell,

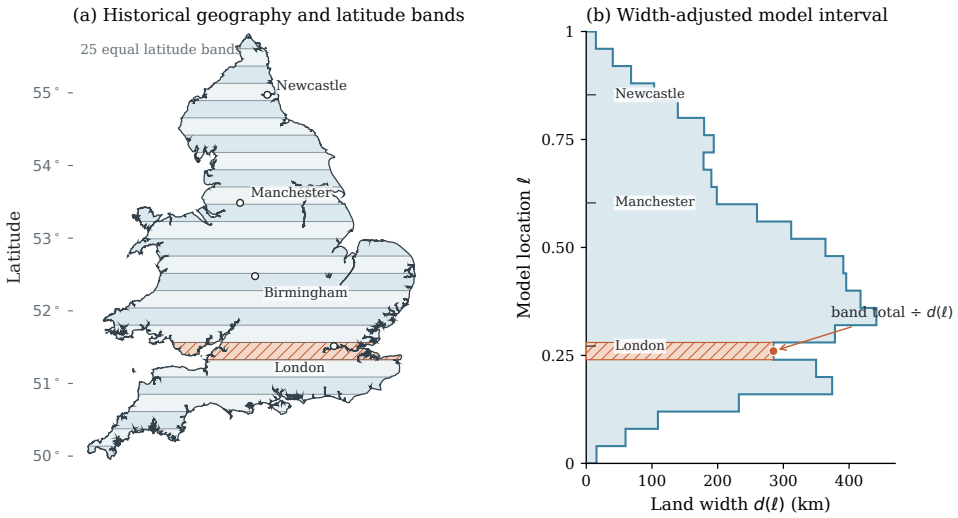


Figure 6: From Historical Geography to the Model Interval

Notes: Panel (a) partitions England and Wales into 25 equal-latitude bands. Panel (b) divides band totals by east–west land width $d(\ell)$ to form densities on $\ell \in [0, 1]$.

1988) in band h and year y, and d_h the band's mean east–west land width, the measured transport stock is

$$\mathcal{I}^T_{hy} = \frac{(M^r_{hy} + 10M^w_{hy} + 50M^g_{hy})/d_h}{25^{-1}\sum_{k=1}^{25}(M^r_{k,1710} + 10M^w_{k,1710} + 50M^g_{k,1710})/d_k}, \tag{75}$$

where the service-capacity weights $(1, 10, 50)$ follow Trew (2020) and the 1710 cross-band mean is normalized to unity. The 1710 cross-section initializes $T_0(\ell)$, while subsequent stocks are generated endogenously by county investment (52)–(53).

Following Hornbeck and Rotemberg (2024), the manufacturing operating wedge $\psi_M(\ell) = \bar{\psi}_M \omega_M(\ell)$ combines a scale parameter $\bar{\psi}_M$ with a predetermined spatial exposure profile $\omega_M(\ell)$ from geocoded fire-insurance registers (Study 1838), normalized to unit 1710 manufacturing-labor-weighted mean ($\frac{\int_0^1 \omega_M(\ell)\omega_{M0}(\ell)\,d\ell}{\int_0^1 \omega_{M0}(\ell)\,d\ell} = 1$ with $\omega_{M0}(\ell) \equiv e_{M,1710}(\ell)L_{M,1710}(\ell)$). By the conditional hiring rule (29), an active manufacturing site with payroll $w(\ell)L_M(\ell)$ and expected gross operating surplus $\mathbb{E}_t[F_{M\ell}] = \mathbb{E}_t\left[[p_M(\ell) - r_M\tau_{j(\ell)}]_+ Y_M(\ell)\right] - w(\ell)L_M(\ell)$ satisfies

$$1 + \psi_M(\ell) = \alpha_M\left(1 + \frac{\mathbb{E}_t[F_{M\ell}]}{w(\ell)L_M(\ell)}\right), \qquad \text{or} \qquad \frac{\alpha_M}{1-\alpha_M}\frac{\mathbb{E}_t[F_{M\ell}]}{w(\ell)L_M(\ell)} = 1 + \frac{\psi_M(\ell)}{1-\alpha_M}, \tag{76}$$

directly calibrating $\bar{\psi}_M = 0.251$ (SE $= 0.032$) from the Clark (2002) nonagricultural labor share, with digitized firm operating accounts (Clough records) providing an external check on $\mathbb{E}_t[F_{M\ell}]/[w(\ell)L_M(\ell)]$.

C Estimation and Identification

The estimation strategy is the Simulated Method of Moments (Bordalo et al., 2026; Hansen, 1982), which jointly estimates $p = 26$ free structural parameters under diagnostic expectations ($\theta \geq 0$ plus the 16 preference, innovation, spatial diffusion, transport, and construction-cost parameters and the 9 sectoral shock parameters in Table 5), while calibrating the manufacturing wedge scale

$\bar{\psi}_M = 0.251$ (SE $= 0.032$) from (76) and concentrating out the eight shape parameters of the five-knot initial 1710 productivity profiles $A_A(\ell, 1710)$ and $A_M(\ell, 1710)$ ($\int_0^1 A_i(\ell, 1710)\, d\ell = 1$) on the 1710 occupational cross-section. To evaluate belief extrapolation, the model is also estimated under the rational expectations restriction ($\theta = 0$, $p = 25$ free SMM parameters). Both specifications share the fixed parameters in Table 3 and minimize the weighted distance between the $M = 30$ historical moments and simulated moments averaged across B common primitive shock histories $\{u_b\}_{b=1}^B$:

$$Q(\Theta) = \big[\overline{\mathbf{m}}(\Theta) - \mathbf{m}^{\text{data}}\big]' W \big[\overline{\mathbf{m}}(\Theta) - \mathbf{m}^{\text{data}}\big], \qquad \overline{\mathbf{m}}(\Theta) = \frac{1}{B}\sum_{b=1}^{B} \mathbf{m}(\Theta; u_b), \tag{77}$$

where $\mathbf{m}^{\text{data}} \in \mathbb{R}^{30}$ stacks the aggregate, transport-network, spatial-covariance, sectoral-cycle, and investment-dynamics moments in Panels A–E of Table 6, and $W = \text{diag}(1/\widehat{\text{SE}}_1^2, \ldots, 1/\widehat{\text{SE}}_{30}^2)$ weights each residual by its inverse empirical variance so that $Q(\Theta) = \sum_{k=1}^{30}\big[(\overline{m}_k(\Theta) - m_k^{\text{data}})/\widehat{\text{SE}}_k\big]^2$ is measured in studentized units (Online Appendix S.D and Table S.D.2 report the formula, standard error $\widehat{\text{SE}}_k$, and weight W_{kk} for each moment).

Table 3: Externally Fixed Parameters and Normalizations

Parameter	Value	Description
β	1/1.05	Annual discount factor (real interest rate of 5%)
$\alpha_A,\ \alpha_M$	0.390, 0.710	Production labor elasticities (agriculture, manufacturing)
r_M	1.000	Manufacturing distribution requirement (normalization)
c	3	Truncation bound for aggregate productivity innovations (in s.d.)
d_0	0.100	Spatial Ornstein–Uhlenbeck correlation length for discovery draws
g_m	log(1.02)	Post-1881 growth rate of physical project increments
$\bar{L}_{1710}$	100	Initial aggregate labor supply (normalization)
$N,\ J$	500, 10	Number of spatial grid cells and county jurisdictions

Notes: Numeraire $p_A(1, t) = 1$; initial productivity profiles satisfy $\int_0^1 A_{i,1710}(\ell)\, d\ell = 1$ ($i \in \{A, M\}$); and the 1710 transport stock has cross-band mean one. Production labor elasticities (α_A, α_M) follow sectoral value-added shares (Herrendorf et al., 2015) (Proposition S.E.2), and demographic mass $\bar{L}_t$ follows (8).

Table 4 summarizes how the moments discipline each parameter block. By Proposition 2, the allocation at each date depends on θ only through county construction, so within-period margins discipline the static parameters given the network, while network dynamics discipline beliefs and construction costs. Sectoral labor shares map into the fixed production elasticities (α_A, α_M), and the nonagricultural labor share, given fire-insurance exposure, calibrates $\bar{\psi}_M$ (76). The 1710 occupational shares, the relative price path, and dated spatial employment covariances discipline (η, γ), $(\alpha_T, \kappa_0, r_A)$, and $A_i(\ell, 1710)$. Aggregate output growth and the shift from agricultural to industrial output identify the innovation and diffusion parameters $(a_i, f_i, \chi_i, \omega_E, \delta, \lambda_0)$, while HP-filtered sectoral output variances, autocovariances, and cross-covariances pin down $(\boldsymbol{\rho}, \boldsymbol{\Sigma})$. Finally, the linear and quadratic trends and subperiod growth rates of log transport capacity (Panel B of Table 6) and the investment-cycle regressions (Panel E) separate θ from $(m_I, \chi_I, \boldsymbol{\rho})$. Under $\theta = 0$, lowering χ_I or raising $\boldsymbol{\rho}$ increases investment–output comovement only by raising the first-order autocorrelation ρ_1 of cyclical investment, whereas $\theta > 0$ amplifies boom-time marginal q, raises 1830–1850 transport growth, and lowers ρ_1 as transitory news subsides. These sign restrictions are verified numerically at the estimates.[8]

[8]Online Appendix S.E states the local rank conditions (Andrews et al., 2017; Rothenberg, 1971) and sign restric-

Table 4: Parameter Blocks and Identifying Moments

Parameter block	Parameters	Primary identifying moments
Preferences	$\eta,\ \gamma$	Linear trends in sectoral employment shares and log relative price $\ln(p_M/p_A)$
Operating wedge	$\bar{\psi}_M$	Nonagricultural labor share and fire-insurance exposure (76)
Innovation	$a_i,\ f_i,\ \chi_i,\ \omega_E$	Linear trends in log real GDP and $\ln(Y_M/Y_A)$; manufacturing share–energy covariance $C_{e,y}$
Spatial diffusion	$\delta,\ \lambda_0$	Manufacturing share–latitude covariance $C_{l,y}$ and spatial autocovariance $C_{N,y}$
Transport	$\alpha_T,\ \kappa_0,\ r_A$	Tertiary employment share trend, cross-band variance $V^{\text{occ}}_{T,y}$, and $\ln(p_M/p_A)$ trend
Construction costs	$m_I,\ \chi_I$	Linear and quadratic trends of log transport stock $\ln \overline{T}_t$; cross-band variance $V_{T,t}$
Diagnostic beliefs	θ	Subperiod transport growth (1830–1850), post-1850 slope change, investment–output slopes (b_0, b_1), autocorrelation (ρ_1), and county lag slopes (g_1, g_2)
Initial productivity	$A_i(\ell, 1710)$	1710 occupational shares and cross-sectional employment covariances
Productivity process	$\boldsymbol{\rho},\ \boldsymbol{\Sigma}$	Variances, first-order autocovariances, and cross-covariances of HP-filtered log sectoral output

Notes: Primary identifying moments for each parameter block (Table 6 and Online Appendix S.D); all blocks are estimated jointly via (77) under DE ($\theta \geq 0$) and RE ($\theta = 0$). Online Appendix S.E states the local rank and sign restrictions.

Following Bordalo et al. (2026), Table 5 reports the parameter estimates and standard errors under diagnostic (DE) and rational (RE, $\theta = 0$) expectations, and Table 6 compares the empirical and simulated moments across all five blocks. Diagnosticity is estimated at $\widehat{\theta} = 0.739$ (SE $= 0.090$), rejecting $\theta = 0$ ($t = 8.21$, $p < 0.0001$). This estimate lies within the benchmark range across financial and macroeconomic settings (Bordalo et al., 2026): slightly below the $\theta \approx 0.8$–0.9 estimated from earnings expectations and credit spreads (Bordalo et al., 2018, 2019), above the $\theta \approx 0.5$–0.6 from macroeconomic surveys (Bordalo et al., 2020), and aligned with the $\theta \approx 0.74$–1.08 in structural models of investment and credit cycles (Bianchi et al., 2024; Bordalo et al., 2026; Maxted, 2023). When the remaining $p = 25$ parameters are re-estimated under $\theta = 0$, the restricted RE estimator lowers adjustment convexity ($\widehat{\chi}_I = 1.579$ versus 1.920) and construction scale ($\widehat{m}_I = 0.020$ versus 0.023), raises manufacturing persistence ($\widehat{\rho}_M = 0.561$ versus 0.533) and volatility ($\widehat{\sigma}_M = 0.042$ versus 0.039), and adjusts spatial catch-up and transport costs ($\widehat{\lambda}_0 = 0.175$ versus 0.160, $\widehat{\kappa}_0 = 0.073$ versus 0.080, $\widehat{\delta} = 1.245$ versus 1.350) to compensate for the absence of extrapolation.

Because those parameter shifts distort other margins, the fit improvement under DE spans Table 6. Both models match long-run output and employment trends (Panel A) and dated spatial sorting covariances (Panel C). The restricted RE model, however, underpredicts the linear and quadratic trends of the log transport stock (1.758 versus 1.948 percent per year and 2.144 versus 2.923 in Panel B), the 1830–1850 railway boom (4.801 versus 7.410 percent per year under DE and 7.470 in the data), and the post-1850 slowdown (-2.461 versus -5.060 and -5.101). Raising $\widehat{\rho}_M$ and $\widehat{\sigma}_M$ under RE also overstates manufacturing and transport cycle variances ($v_M = 12.12$ versus 9.96 under DE and 9.64 in the data; $v_T = 7.37$ versus 5.86 and 5.66 in Panel D), shifts investment comovement from current to lagged output ($b_0 = 2.063$ and $b_1 = 3.142$ under RE, versus 3.091 and 1.662 under DE and 2.954 and 1.585 in the data), and mutes both the negative serial correlation of

tions; Replication Appendix R.A details the sequence-space Jacobian evaluation of (77) (Auclert et al., 2021).

Table 5: SMM Parameter Estimates: Diagnostic versus Rational Expectations

Parameter	Description	DE	RE ($\theta = 0$)
Panel A. Preferences, production, and trade costs			
η	Stone–Geary food expenditure share	0.520 (0.028)	0.540 (0.031)
γ	Subsistence food requirement	0.042 (0.003)	0.039 (0.003)
α_A	Agricultural labor elasticity	0.390 [fixed]	0.390 [fixed]
α_M	Manufacturing labor elasticity	0.710 [fixed]	0.710 [fixed]
$\bar{\psi}_M$	Manufacturing labor wedge scale	0.251 (0.032) [calibrated]	0.251 (0.032) [calibrated]
α_T	Transport labor elasticity	0.450 (0.029)	0.473 (0.033)
κ_0	Iceberg transport cost scale	0.080 (0.007)	0.073 (0.007)
r_A	Agricultural distribution requirement ($r_M = 1$)	0.200 (0.015)	0.216 (0.018)
Panel B. Innovation and spatial diffusion			
a_A	Agricultural innovation Pareto shape	2.200 (0.150)	2.122 (0.156)
a_M	Manufacturing innovation Pareto shape	1.800 (0.112)	1.887 (0.126)
f_A	Agricultural fixed innovation cost	0.040 (0.004)	0.043 (0.005)
f_M	Manufacturing fixed innovation cost	0.030 (0.003)	0.028 (0.003)
χ_A	Agricultural variable innovation cost	0.080 (0.008)	0.085 (0.009)
χ_M	Manufacturing variable innovation cost	0.060 (0.006)	0.055 (0.006)
ω_E	Coalfield innovation cost loading	0.0040 (0.0006)	0.0045 (0.0007)
λ_0	Baseline technological catch-up rate	0.160 (0.014)	0.175 (0.016)
δ	Spatial knowledge decay rate	1.350 (0.097)	1.245 (0.097)
Panel C. Infrastructure investment and expectations			
θ	Diagnostic expectations parameter	0.739 (0.090)	0 [fixed]
m_I	Infrastructure construction cost scale	0.023 (0.006)	0.020 (0.005)
χ_I	Infrastructure adjustment cost convexity	1.920 (0.233)	1.579 (0.213)
Panel D. Aggregate productivity process			
ρ_A	AR(1) persistence: Agriculture	0.168 (0.042)	0.174 (0.043)
ρ_M	AR(1) persistence: Manufacturing	0.533 (0.124)	0.561 (0.108)
ρ_T	AR(1) persistence: Transport	0.303 (0.076)	0.318 (0.079)
σ_A	Innovation s.d.: Agriculture	0.067 (0.004)	0.070 (0.004)
σ_M	Innovation s.d.: Manufacturing	0.039 (0.002)	0.042 (0.002)
σ_T	Innovation s.d.: Transport	0.023 (0.002)	0.025 (0.002)
ϱ_{AM}	Innovation correlation: (A, M)	−0.058 (0.015)	−0.082 (0.020)
ϱ_{AT}	Innovation correlation: (A, T)	0.094 (0.024)	0.077 (0.019)
$\varrho_{MT\|A}$	Partial innovation correlation: $(M, T \mid A)$	0.337 (0.084)	0.373 (0.093)

Notes: SMM estimates minimizing (77) across $M = 30$ moments (Table 6) under diagnostic (DE, $\theta \geq 0$, $p = 26$) and rational (RE, $\theta = 0$, $p = 25$) expectations across $B = 1{,}000$ common shock histories, with misspecification-robust sandwich standard errors (S.D.11) in parentheses. The wedge scale $\bar{\psi}_M = 0.251$ (SE = 0.032) is calibrated via (76), the eight 1710 productivity spline coefficients are concentrated out on the 1710 occupational cross-section, and fixed parameters are in Table 3.

investment ($\rho_1 = -0.183$ versus -0.344 under DE and -0.325 in the data) and county investment reversals after population growth ($g_1 = -0.082$ and $g_2 = -0.144$ under RE, versus -0.190 and -0.484 under DE and -0.162 and -0.496 in the data; Panel E). Online Appendix Table S.E.1 confirms that this fit gain does not stem from an extra parameter or moment weighting: the Rivers and Vuong (2002) equal-parameter test ($p = 25$ free parameters in every column) favors $\theta = 0.9$ ($Q = 5.54$) over $\theta = 0$ ($Q = 117.86$) with $z = 4.35$ ($p < 0.0001$). The Andrews et al. (2017) sensitivity measures show that Panels E and B supply 85.7 percent (49.8 and 35.9 percent) of the identifying variation for $\widehat{\theta}$, while leave-one-block-out estimations yield $\widehat{\theta} \in [0.666, 0.747]$ and equal-block studentized weights yield $\widehat{\theta}_{\text{eq}} = 0.748$ (SE = 0.104).

Table 6: Empirical Moments and Model Fit

Empirical moment	Window	Data	DE	RE
Panel A. Aggregate growth and structural transformation				
Log real GDP trend (%/yr)	1710–1881	1.473 (0.210)	1.425	1.364
Log relative output $\ln(Y_M/Y_A)$ trend (%/yr)	1710–1881	1.085 (0.572)	0.970	0.799
Log relative price $\ln(p_M/p_A)$ trend (%/yr)	1710–1870	−0.757 (0.616)	−0.632	−0.496
Agricultural employment share trend (pp/yr)	1710–1881	−0.146 (0.023)	−0.139	−0.116
Tertiary employment share trend (pp/yr)	1710–1881	0.060 (0.007)	0.058	0.055
Panel B. Transport network growth and spatial dispersion				
Log transport stock: linear trend (%/yr)	1710–1881	1.948 (0.103)	1.943	1.758
Log transport stock: quadratic trend (×100)	1710–1881	2.923 (0.217)	2.952	2.144
Log transport stock: 1830–1850 growth (%/yr)	1830–1850	7.470 (0.759)	7.410	4.801
Log transport stock: post-1850 slope change (%/yr)	1830–1881	−5.101 (0.881)	−5.060	−2.461
Cross-band variance of log transport stock	1710–1881	0.469 (0.148)	0.437	0.410
Panel C. Cross-sectional employment covariances				
Manufacturing share–latitude covariance	1710–1881	0.077 (0.026)	0.070	0.065
Manufacturing share–energy covariance	1710–1881	−0.070 (0.024)	−0.064	−0.059
Spatial autocovariance of manufacturing share	1710–1881	0.030 (0.006)	0.028	0.026
Cross-band variance of tertiary share	1710–1881	0.0029 (0.0011)	0.0027	0.0025
Panel D. HP-filtered log sectoral output moments ($\times 10^4$)				
Variance: Agriculture (A)	1710–1881	34.79 (4.60)	35.81	39.80
Variance: Manufacturing (M)	1710–1881	9.64 (1.40)	9.96	12.12
Variance: Transport (T)	1710–1881	5.66 (0.93)	5.86	7.37
First-order autocovariance: Agriculture (A)	1710–1881	−7.27 (2.41)	−6.86	−6.75
First-order autocovariance: Manufacturing (M)	1710–1881	−0.49 (0.68)	−0.35	−0.08
First-order autocovariance: Transport (T)	1710–1881	−0.63 (0.39)	−0.55	−0.39
Cross-sector covariance: (A, M)	1710–1881	−2.59 (1.43)	−2.89	−3.89
Cross-sector covariance: (A, T)	1710–1881	−0.55 (0.83)	−0.71	−1.20
Cross-sector covariance: (M, T)	1710–1881	4.74 (0.78)	4.95	6.34
Panel E. Infrastructure investment dynamics				
Standard deviation of cyclical investment, $\sigma(c_t^I)$	1710–1880	0.431 (0.043)	0.427	0.345
Investment on current output cycle (b_0)	1711–1880	2.954 (1.431)	3.091	2.063
Investment on lagged output cycle (b_1)	1711–1880	1.585 (1.257)	1.662	3.142
First-order autocorrelation of investment (ρ_1)	1711–1880	−0.325 (0.074)	−0.344	−0.183
Lead output cycle on investment (b_{lead})	1711–1880	−0.0051 (0.0039)	−0.0049	−0.0035
County investment on population growth, lag 1 (g_1)	1791–1851	−0.162 (0.050)	−0.190	−0.082
County investment on population growth, lag 2 (g_2)	1791–1851	−0.496 (0.050)	−0.484	−0.144

Notes: Empirical moments (with source-bootstrap or Newey–West HAC standard errors in parentheses; Online Appendix S.D.8 and Table S.D.2) and simulated means across $B = 1{,}000$ common shock histories at the DE and RE estimates in Table 5. Sources: Broadberry et al. (2022) and Feinstein (1972) (Panels A, D), Shaw-Taylor et al. (2019) and Keibek (2017) (Panels A, C), and the measured transport stock (75) (Panels B, E).

D Aggregate Takeoff and Spatial Reallocation

Figures 7 and 8 plot the historical transition alongside the mean simulated trajectories and 5th–95th percentile bands across 1,000 Monte Carlo draws under DE and RE. Total real GDP rose 14.33-fold between 1710 and 1881, averaging 1.56 percent per year (Figure 7, Panel a), as annual growth accelerated from 1.02 percent in 1710–1760 and 1.16 percent in 1760–1800 to 1.60 percent in 1800–1830 and 2.37 percent in 1830–1881. Because no moment of the real wage series enters the SMM objective in Table 6, Panel (b) of Figure 7 provides an untargeted check on the model's price and wage predictions. Historically, real wages $w/\mathcal{P}$ rose modestly in the early eighteenth century, plateaued at 1.35–1.55 times their 1710 level during the late eighteenth and early nineteenth centuries—the "Engels' pause" of Allen (2009)—and then accelerated after 1820 to 3.28 times their 1710 level by 1881 as transport integration and industrial productivity gains lowered delivered prices. The industrial-to-agricultural output ratio Y_M/Y_A (Panel c) follows the same two-phase

timing, drifting to 0.60 times its 1710 level in 1781 as population growth raised food demand before accelerating at 2.61 percent per year after 1813 to 5.77 times its 1710 level by 1881.[9] The estimated DE model reproduces this two-phase acceleration in aggregate output, Y_M/Y_A, and the untargeted real wage path $w/\mathcal{P}$ (1.41 in 1760, 1.75 in 1820, and 3.11 in 1881), whereas the restricted RE trajectory matches the eighteenth-century plateau (1.35 in 1760 and 1.47 in 1820) but stalls at 1.72 by 1881 alongside a muted rise in Y_M/Y_A.

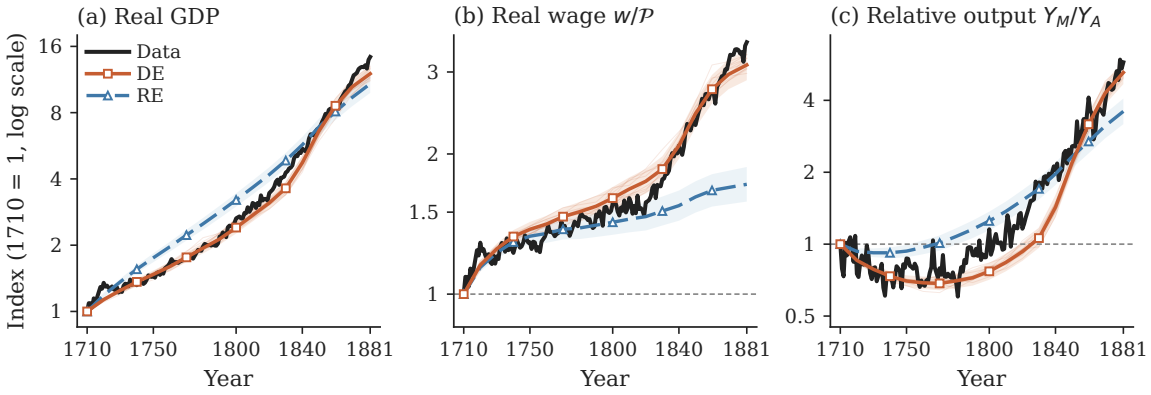


Figure 7: Aggregate Transition, 1710–1881

Notes: Historical series and simulated means (with 5th–95th percentile bands across 1,000 draws) under diagnostic (DE, $\widehat{\theta} = 0.739$) and restricted rational (RE, $\theta = 0$) expectations (1710 = 1, log scale). Panels (a) and (c) plot real GDP and relative industrial output Y_M/Y_A (Broadberry et al., 2022; Feinstein, 1972); Panel (b) plots the untargeted aggregate real wage $w/\mathcal{P}$ (Broadberry et al., 2022; Clark, 2002).

Transport capacity expanded nonlinearly over the sample. The measured stock (75) rose 27.70-fold between 1710 and 1881 (Figure 8, Panel c). Before 1830 it grew at 0.99 percent per year to 3.28 times its 1710 level, with turnpike-era bursts of 1.79 percent per year in 1750–1770 and 1.68 percent in the 1790s. Railway construction raised annual log growth to 5.90 percent in the 1830s and 8.43 percent in the 1840s (7.47 percent per year over 1830–1850), lifting the stock to 5.91 by 1840 and 13.73 by 1850, before annual growth slowed by 5.10 percentage points to 2.37 percent over 1850–1881 (24.36 in 1870). Disciplined by Panel B of Table 6, the DE path tracks both gradual eighteenth-century accumulation and the 1830–1850 surge and post-1850 moderation, whereas the RE path grows too smoothly and misses the mid-century acceleration. The 1881 band profile (Panel d) is multi-peaked: the West Riding and Lancashire band carried 53.8 times the 1710 national mean, followed by London (46.0) and the Midlands–Northern industrial bands (35.1–40.1), while Cornwall (3.5) and Northumberland (8.2) remained least served. Cross-band log-stock variance $V_{T,y}$ fell from 0.713 in 1710 to 0.295 in 1881 in the data (0.658 to 0.240 under DE; Figure S.D.1, Panel f), with a sample mean of $\overline{V}_T = 0.469$ (0.437 under DE and 0.410 under RE; Panel B). Under DE, counties build more aggressively during the canal and railway booms across the Midlands and Northern corridor (bands 12–22) than under RE, matching the 1881 cross-sectional peaks.

Panels (a)–(b) of Figure 8 compare the 1817–1881 change in manufacturing and tertiary employment shares across the twenty-five latitude bands with the DE and RE predictions (Online Appendix S.B). As the national manufacturing share rose from 49.9 to 62.1 percent, industrial employment diffused across space: the manufacturing share rose in 20 of 25 bands and its change correlates negatively with the 1817 share (−0.81), with 20–26 percentage point gains in initially

[9]The relative price p_M/p_A (Online Appendix Figure S.D.1, Panel c) mirrors this shift inversely, remaining above its 1710 level through much of the eighteenth century (peaking at 1.47 in 1733) before falling below unity after 1799 to 0.71 in 1800, 0.59 in 1830, 0.39 in 1855, and 0.49 in 1870.

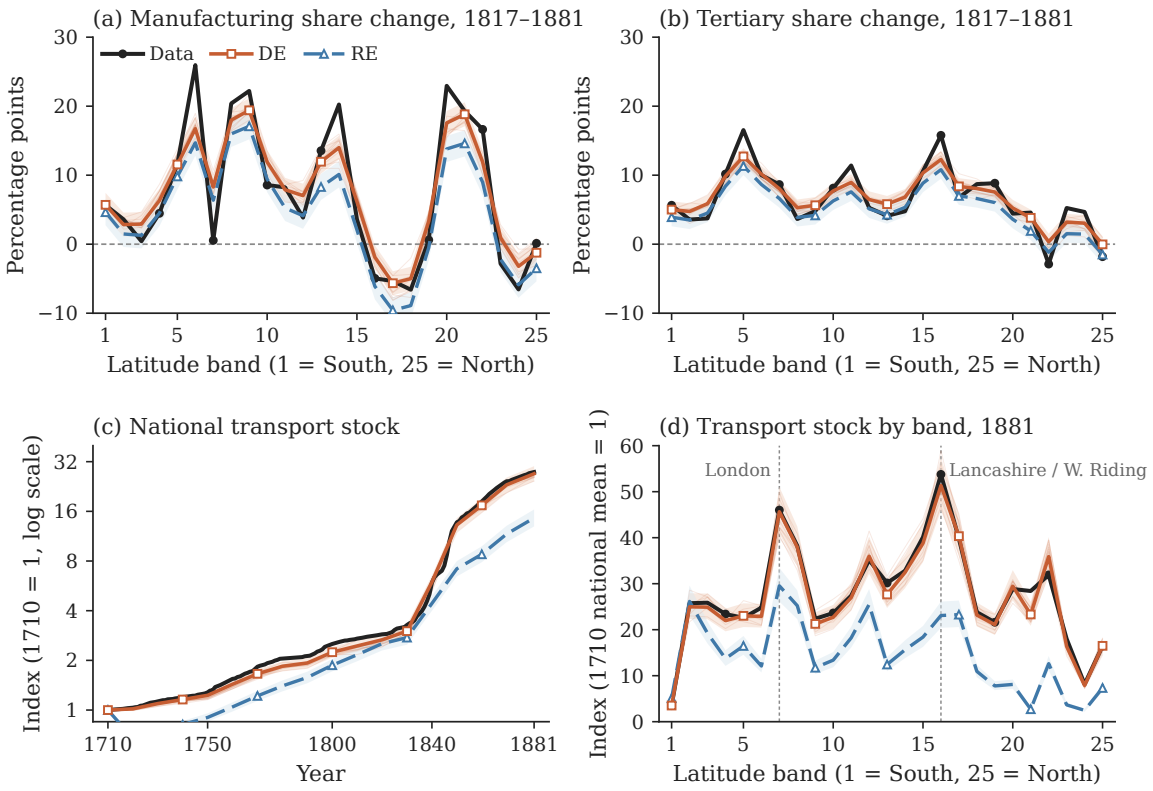


Figure 8: Spatial Reallocation and Infrastructure, 1710–1881

Notes: Historical series and simulated means (with 5th–95th percentile bands across 1,000 draws) under DE ($\widehat{\theta} = 0.739$) and RE ($\theta = 0$). Panels (a)–(b) plot 1817–1881 percentage-point changes in manufacturing and tertiary employment shares across the 25 latitude bands (1 = South, 25 = North; Online Appendix S.B); Panels (c)–(d) plot the national mean transport stock (75) (1710 = 1, log scale) and the 1881 band profile (1710 national mean = 1).

peripheral bands (bands 6, 8, and 20–22) and 5–7 point declines in the Lancashire–Yorkshire core (bands 16–18, which remained above 79 percent manufacturing in 1881). Tertiary employment expanded in 23 of 25 bands as the national transport and distribution share nearly doubled from 9.5 to 18.5 percent, led by the Southern, London, and Lancashire–West Riding corridors. Although the estimation targets only aggregate occupational series and dated spatial covariances in Panels A and C, the DE model captures both manufacturing diffusion and tertiary expansion more closely than RE.[10]

E Counterfactual Analysis: Diagnostic versus Rational Expectations

Re-estimating the model under $\theta = 0$ alters technology and construction-cost parameters alongside beliefs. To isolate diagnostic expectations, the belief-neutralization counterfactual (R | DE) holds all other parameters at their diagnostic estimates, together with the 1710 state, statutory project opportunities, and primitive shock draws $\{u_b\}_{b=1}^{B}$, sets $\theta = 0$, and recomputes county investment policies $\mathbf{x}_t^0$ and the general equilibrium path from 1710 to 1881.

By Proposition 2, household and firm decisions and prices at each date are invariant to θ given county investment, so every difference between the DE and R | DE paths originates in county construction and propagates through the installed network. Construction absorbs manufactures

[10]Online Appendix S.D (Figure S.D.1) plots the complete historical and simulated series for every moment block.

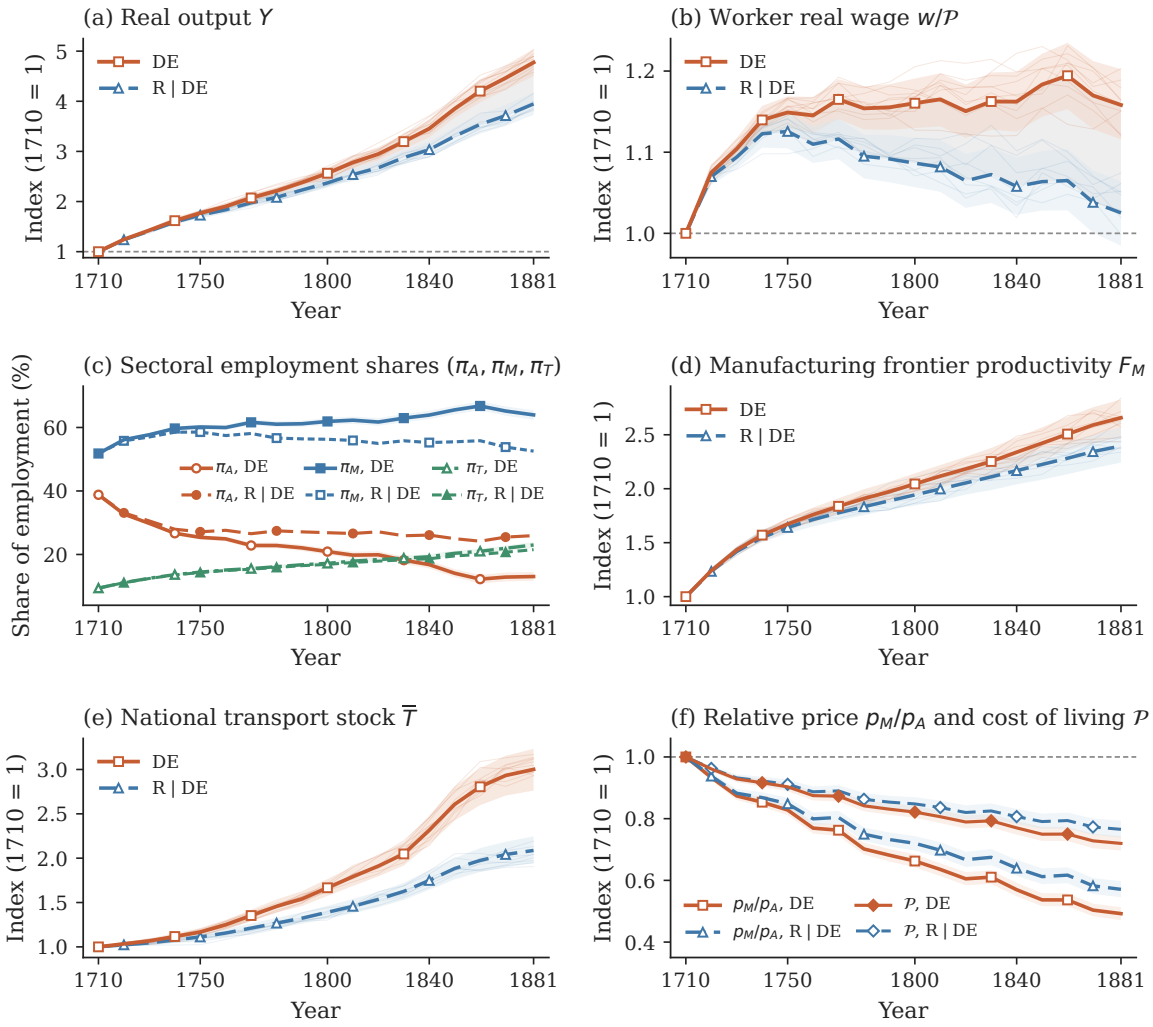


Figure 9: Counterfactual Effects of Diagnostic Expectations, 1710–1881

Notes: Simulated means and 5th–95th percentile bands (1,000 draws) under the baseline diagnostic economy (DE, $\widehat{\theta} = 0.739$) and the belief-neutralization counterfactual (R | DE: $\theta = 0$ holding all other parameters and shock histories fixed). Panels (a), (b), (d), (e), and (f) report balanced-growth-detrended indices (1710 = 1) for real output Y, real wage $w/\mathcal{P}$, manufacturing frontier F_M, transport stock $\overline{T}$, and prices ($p_M/p_A, \mathcal{P}$); Panel (c) reports sectoral employment shares π_A, π_M, π_T (percent).

when built; completed capacity lowers route losses $\kappa(T)$ and expands carrier service capacity at later dates; and, by Proposition 3, higher capacity accelerates spatial knowledge catch-up.

Figure 9 and Table 7 report the counterfactual.[11] By 1881, diagnostic expectations expand the aggregate transport stock $\overline{T}$ by +43.79 percent (3.001 versus 2.087; Panel e), with regional gains of +29.53 percent in the Southern belt (Counties 1–5), +54.47 percent in the Midlands–North corridor (Counties 6–9), and +61.33 percent in the Northern border (County 10). This network expansion raises 1881 real output Y (and output per worker y) by +20.96 percent (4.774 versus 3.947; Panel a) and lifts the population-weighted real wage $w/\mathcal{P}$ by +12.97 percent nationally (1.158 versus 1.025; +12.78 to +13.59 percent across regions; Panel b). Larger route capacity accelerates structural

[11] Level indices in Figure 9 and Table 7 are in balanced-growth-detrended units (1710 = 1), and employment shares are model-domain shares $\mathcal{L}_o(t)/\bar{L}_t$ (S.D.2). Because exogenous package-scale and frontier drift factors match across DE and R | DE, percentage differences 100 (DE/R | DE − 1) are identical in un-detrended units.

transformation (Trew, 2020), lowering the 1881 agricultural employment share by −12.9 percentage points (13.0 versus 25.9 percent) and shifting +11.4 percentage points into manufacturing (64.0 versus 52.6 percent) and +1.5 percentage points into tertiary services (23.0 versus 21.5 percent; Panel c). Faster spatial diffusion lifts manufacturing frontier productivity F_M by +10.90 percent (2.657 versus 2.396; Panel d), while cheaper distribution and higher manufacturing productivity lower p_M/p_A by −13.79 percent (0.492 versus 0.571) and the Stone–Geary price index $\mathcal{P}$ by −5.97 percent (0.720 versus 0.765; Panel f).

Table 7: Counterfactual Effects of Diagnostic Expectations in 1881

1881 outcome (1710 = 1, except shares)	DE ($\widehat{\theta} = 0.739$)	R | DE ($\theta = 0$)	Difference
Panel A. Aggregate outcomes			
Real output (Y)	4.774	3.947	+20.96%
Population-weighted real wage ($w/\mathcal{P}$)	1.158	1.025	+12.97%
Agricultural employment share (π_A, %)	13.0	25.9	−12.9 pp
Manufacturing / tertiary shares (π_M/π_T, %)	64.0 / 23.0	52.6 / 21.5	+11.4 / +1.5 pp
Manufacturing frontier productivity (F_M)	2.657	2.396	+10.90%
Aggregate transport stock ($\overline{T}$)	3.001	2.087	+43.79%
Relative price of manufactures (p_M/p_A)	0.492	0.571	−13.79%
Stone–Geary price index ($\mathcal{P}$)	0.720	0.765	−5.97%
Panel B. Regional outcomes ($w/\mathcal{P}$ / F_M / $\overline{T}$)			
Southern belt (Counties 1–5)	1.188 / 1.888 / 2.470	1.053 / 1.703 / 1.907	+12.78% / +10.90% / +29.53%
Midlands–North corridor (Counties 6–9)	1.156 / 3.299 / 3.549	1.023 / 2.975 / 2.298	+13.00% / +10.90% / +54.47%
Northern border (County 10)	1.116 / 3.935 / 3.463	0.982 / 3.548 / 2.146	+13.59% / +10.90% / +61.33%

Notes: 1881 outcomes averaged across $B = 1{,}000$ common shock histories under baseline diagnostic expectations (DE, $\widehat{\theta} = 0.739$) and the belief-neutralization counterfactual (R | DE, $\theta = 0$ holding all other parameters and shock histories fixed). Level indices are in balanced-growth-detrended units (1710 = 1; percentage differences $100\,(\text{DE}/\text{R} \mid \text{DE} - 1)$ are identical in historical units), and employment shares are in percent (differences in percentage points).

Two features of Table 7 highlight the model's spatial mechanisms. First, in Panel B, the counterfactual percentage gain in the available frontier $F_{Mt}(\ell) = \sup_{r\in[0,1]} e^{-\delta|\ell-r|} A^{+}_{Mt}(r)$ is spatially uniform at +10.90 percent across all three regions because the spatial supremum scales proportionally with the northern industrial peak. Local applied manufacturing productivity $A_{Mt}(\ell)$, governed by the transport-dependent catch-up rate $\lambda_t(\ell) = \lambda_0[T_t(\ell)/T_{t_0}(\ell)]^{1-\alpha_T}$ in (39)–(40), expands most where route buildout is largest (+7.84 percent in the Southern belt, +13.42 percent in the Midlands–North corridor, and +14.18 percent in the Northern border, or +11.65 percent nationally). Second, decomposing the +20.96 percent 1881 output gain sequentially across the three uninternalized social return channels of Section III.F shows why static "social savings" calculations (Fogel, 1964) capture only a fraction of the long-run return to connective infrastructure: direct carrier and iceberg transit savings holding spatial sorting and technology at their R | DE paths (Donaldson and Hornbeck, 2016; Fogel, 1964) account for +5.42 percentage points (25.9 percent of the gain); spatial labor reallocation into wedge-distorted manufacturing ($\psi_M(\ell) > 0$; Baqaee and Farhi 2020; Hornbeck and Rotemberg 2024; Hsieh and Klenow 2009) accounts for +6.18 percentage points (29.5 percent); and transport-mediated spatial knowledge diffusion and induced innovation (65) (Desmet and Rossi-Hansberg, 2014; Mokyr, 2005; Nelson and Phelps, 1966; Romer, 1990) account for +9.36 percentage points (44.7 percent).

V Conclusion

Economic historians and financial economists have long viewed the recurrent investment manias of the British Industrial Revolution—and subsequent general-purpose network buildouts from electrification to fiber-optic telecommunications (Bresnahan and Trajtenberg, 1995)—through two seemingly incompatible lenses. Financial histories emphasize promoter over-extrapolation, post-boom construction reversals, and disappointing realized yields (Arnold and McCartney, 2011; Campbell, 2012; Keynes, 1936; Kindleberger, 1978; Odlyzko, 2010; Shaen, 1847; Shiller, 2000); growth and spatial histories emphasize that the physical networks left behind permanently integrated national markets and accelerated industrialization (Bogart, 2014; Donaldson and Hornbeck, 2016; Fogel, 1964; Mokyr, 2005; Trew, 2020). Embedding a Markov perfect county infrastructure game under diagnostic expectations into a continuous-space endogenous growth model of England and Wales over 1710–1881 shows that both regularities are two sides of a single second-best mechanism: *corrective overoptimism* (Lipsey and Lancaster, 1956). Because decentralized promoters appropriate only local land rents inside their jurisdictions, rational expectations chronically underinvest relative to the social returns from inter-county transit savings, labor reallocation into wedge-distorted manufacturing, and spatial technology diffusion. Diagnostic overreaction to favorable productivity news induces promoters to build aggressively during booms, and physical irreversibility (Arrow, 1968; Dixit and Pindyck, 1994) locks in the resulting route capacity long after expectations revert.

Estimating the model via the Simulated Method of Moments yields a diagnosticity parameter of $\widehat{\theta} = 0.739$ (SE $= 0.090$), matching the $\theta \approx 0.74$–0.91 range documented in financial and macroeconomic expectations (Bordalo et al., 2018, 2026, 2019) and identified by the coexistence of strong boom-time investment comovement with sharp post-boom reversals. In the belief-neutralization counterfactual (R | DE), diagnostic expectations expand the 1881 aggregate transport stock by +43.79 percent (+54.47 percent in the Midlands–North corridor), accelerate structural transformation out of agriculture by −12.9 percentage points, lift manufacturing frontier productivity by +10.90 percent, and raise 1881 real output by +20.96 percent and worker real wages by +12.97 percent.

These findings carry three broader implications. First, they recast the "social savings" debate initiated by Fogel (1964): static freight-cost savings capture only about one-quarter (25.9 percent, or +5.42 percentage points) of the long-run output gain from belief-driven network expansion, while nearly three-quarters arises from reallocating labor into wedge-distorted manufacturing (29.5 percent, or +6.18 percentage points; Baqaee and Farhi 2020; Hornbeck and Rotemberg 2024; Hsieh and Klenow 2009; Restuccia and Rogerson 2008) and accelerating the spatial diffusion of frontier discoveries (44.7 percent, or +9.36 percentage points; Aghion and Howitt 1992; Lucas 1988; Krugman 1991; Mokyr 2005; Nelson and Phelps 1966; Romer 1990). Second, they establish a primitive boundary condition separating *corrective* from *destructive* belief overreaction (Bordalo et al., 2018, 2026, 2020; Kahneman and Tversky, 1972): when extrapolative beliefs fund non-durable consumption or zero-sum speculation without technological externalities, the bust leaves balance-sheet distress with little productive legacy; when overoptimism funds irreversible connective capital whose social returns exceed private appropriation, private financial disappointment coexists with permanent productivity gains. Finally, corrective overoptimism highlights a comparative institutional trade-off (North, 1990): continental European states addressed infrastructure underprovision through centralized planning and sovereign dividend guarantees (Bogart, 2014; Milward and Saul, 1973), whereas Britain and the United States relied on decentralized private promotion amplified by speculative manias (Janeway, 2018; Morck, 2022). Policies that lean uniformly against investor exuberance—or insulate incumbent network providers from entry—risk suppressing the decentralized investment waves that overcome chronic underprovision.

subsection

A Proofs of Main Results

A.1 Proof of Proposition 1

Proof. Because $\mathbb{E}_t^\theta$ is the expectation under the tilted probability in (10), it is linear, so the objective in (13) equals $\mathbb{E}_t^\theta[U(c_A(t), c_M(t))] + \mathbb{E}_t^\theta[\sum_{s\geq t+1} \beta^{s-t} U(c_A(s), c_M(s))]$. Goods are nonstorable, households neither save nor borrow, relocation is costless, portfolio income b_s is residence independent, and households are atomistic, so the set of feasible continuation plans $\{\ell_s, c_A(s), c_M(s)\}_{s\geq t+1}$ does not depend on $(\ell_t, c_A(t), c_M(t))$ and the supremum C_t of the second term is the same for every current residence. Ex-post consumption optimization gives $U(c_A(t), c_M(t)) = u_t(\ell_t, \omega)$ statewise by (15), so $W_t = \sup_{\ell_t \in [0,1]} \mathbb{E}_t^\theta[u_t(\ell_t)] + C_t$, where finiteness of W_t and of every $\mathbb{E}_t^\theta[u_t(\ell)]$ rules out an $\infty - \infty$ ambiguity. Hence occupied locations maximize $\mathbb{E}_t^\theta[u_t(\ell)]$ at the common value $\bar{u}_t$ and unoccupied locations satisfy $\mathbb{E}_t^\theta[u_t(\ell)] \leq \bar{u}_t$. No Bellman recursion is used: diagnostic expectations are not dynamically consistent, so $\mathbb{E}_t^\theta$ applied to the date-$(t+1)$ value formed under $\mathbb{E}_{t+1}^\theta$ need not equal the date-t perceived continuation. [← Back to Proposition 1] □

A.2 Proof of Proposition 2

Proof. Let $\mathcal{F}_t^+$ be the σ-field generated by $(\Omega_t, \mathcal{E}_t, \mathbf{x}_t)$ and the date-t discovery fields $(X_{it}^s, X_{it}^g)_i$. By the selection rule in the statement, the conditional dated equilibrium of Definition 1 is $\mathcal{F}_t^+$-measurable, and $\mathbf{z}_{t+1}$ is independent of $\mathcal{F}_t^+$ conditional on Ω_t because the discovery fields are independent across dates and of the aggregate gaps. In (10) the likelihood ratio $\Lambda_t(\mathbf{z}_{t+1})^\theta$ is a function of $\mathbf{z}_{t+1}$ alone, so for every integrable $\mathcal{F}_t^+$-measurable X_t, $\mathbb{E}_t[X_t \Lambda_t(\mathbf{z}_{t+1})^\theta] = \mathbb{E}_t[X_t]\,\mathbb{E}_t[\Lambda_t(\mathbf{z}_{t+1})^\theta]$ and $\mathbb{E}_t^\theta[X_t] = \mathbb{E}_t[X_t]$. Every within-period objective in Section III—the worker's expected indirect utility (35), the producer's expected net revenue (27), the carrier's expected toll (49), and the county current dividend (55)—is of this form, so the conditional dated equilibrium is invariant to θ once $\mathbf{x}_t$ is fixed, and the next physical state follows from the same discoveries and shocks through (37)–(39) and (53). [← Back to Proposition 2] □

A.3 Proof of Proposition 3

Proof. Let $(\mathcal{D}A^+)(\ell) = \sup_{r\in[0,1]} e^{-\delta|\ell - r|} A^+(r)$ and $F = \mathcal{D}A^+$. Taking $r = \ell$ gives $A^+ \leq F$, and the triangle inequality $|\ell - s| \leq |\ell - r| + |r - s|$ implies $\mathcal{D}F = F$. Because $A' = e^{-\lambda}A^+ + (1 - e^{-\lambda})F$ is a convex combination of A^+ and F for $\lambda \geq 0$, we have $A^+ \leq A' \leq F$ and hence $F = \mathcal{D}A^+ \leq \mathcal{D}A' \leq \mathcal{D}F = F$. Without further discoveries, F is unchanged and iterating $F - A' = e^{-\lambda}(F - A^+)$ gives (41); since $\lambda_t(\ell)$ increases in $T_t(\ell)$ and $F - A^+ \geq 0$, higher $T_t(\ell)$ weakly raises $A_{i,t+1}(\ell)$. [← Back to Proposition 3] □

A.4 Proof of Proposition 4

Proof. Since $\mathcal{X}_t = \prod_j [0, \bar{x}_{jt}]$ is nonempty, compact, and convex and each Π_{jt}^θ is continuous in $\mathbf{x}$ and concave in x_{jt}, Kakutani's fixed-point theorem yields a nonempty compact Nash equilibrium set $\mathcal{N}_t^\theta$. When Π_{jt}^θ is differentiable, concavity makes the variational inequality

$$g_{jt}^\theta(\mathbf{x}^*)(u - x_j^*) \leq 0 \quad \forall\, u \in [0, \bar{x}_{jt}], \quad j = 1, \ldots, J, \qquad g_{jt}^\theta = \partial_j \Pi_{jt}^\theta, \tag{A.1}$$

necessary and sufficient. Under (62), each best response $b_j(\mathbf{y}_{-j})$ is unique, and comparing $u = b_j(\mathbf{y}_{-j})$ with $v = b_j(\mathbf{z}_{-j})$ in (A.1) gives $(u - v)\{g_j(u, \mathbf{y}_{-j}) - g_j(v, \mathbf{z}_{-j})\} \geq 0$, hence $m_j|u - v| \leq$

$\sum_{k\neq j} L_{jk}|y_k - z_k|$. In the weighted maximum norm $\|\mathbf{q}\|_v = \max_j |q_j|/v_j$, we obtain $\|\mathbf{b}(\mathbf{y}) - \mathbf{b}(\mathbf{z})\|_v \leq \gamma_I \|\mathbf{y} - \mathbf{z}\|_v$ with $\gamma_I < 1$ by (63), so Banach's fixed-point theorem yields a unique Nash equilibrium and geometric convergence of best responses. Under strong monotonicity instead, summing the J inequalities (A.1) at two equilibria $\mathbf{x}$ and $\mathbf{y}$ (with $u = y_j$ and $u = x_j$, respectively) gives $\sum_j (x_j - y_j)[g_j(\mathbf{y}) - g_j(\mathbf{x})] \leq 0$, so $\mu\|\mathbf{x} - \mathbf{y}\|^2 \leq 0$ and $\mathbf{x} = \mathbf{y}$ (Vives, 1999). Measurability of the selection in Ω_t follows in the first case because the equilibrium is the pointwise limit of the measurable best-response iterates (each a measurable maximizer of a Carathéodory payoff), and in the second because it is the unique solution of the variational inequality with a Carathéodory pseudogradient. [← Back to Proposition 4] □

A.5 Proof of Proposition 5

Proof. Fix a realization ω with absolutely continuous $p_i(\cdot, \omega)$ and $H_i(\cdot, \omega)$, so that $p_i H_i$ is absolutely continuous and the fundamental theorem of calculus applies to it. Writing local carrier demand as $D = \sum_i r_i(Y_i + |H_i|)$ and using local labor clearing $n = \sum_i e_i L_i + L_T$, summing the local ledgers (69)–(71) gives

$$R^{\text{net}} + \Pi_F + \Pi_T + \Pi_\psi = \sum_i p_i Y_i + \tau_{j(\ell)} \sum_i r_i |H_i| + \tau_{j(\ell)}(Y_T - D) - wn - \sum_i e_i \Psi_i - p_M I. \quad \text{(A.2)}$$

Because τ_j is constant within each county $\mathcal{C}_j$, county service clearing (50) implies

$$\int_{\mathcal{C}_j} \tau_j Y_T \, d\ell = \tau_j K_j = \tau_j D_j = \int_{\mathcal{C}_j} \tau_j D \, d\ell, \quad \text{(A.3)}$$

so integrating (A.2) over $[0, 1]$ yields

$$b_t \bar{L}_t = \int_0^1 (R^{\text{net}} + \Pi_F + \Pi_T + \Pi_\psi) d\ell = \int_0^1 \Big[\sum_i p_i Y_i + \tau_{j(\ell)} \sum_i r_i |H_i| - wn - \sum_i e_i \Psi_i - p_M I\Big] d\ell. \quad \text{(A.4)}$$

Meanwhile, combining goods conservation (68) with $H_i \partial_\ell p_i = (\kappa p_i + r_i \tau_{j(\ell)})|H_i|$ from spatial no-arbitrage (44) cancels the physical loss terms $\pm\kappa p_i |H_i|$ in $\partial_\ell(p_i H_i) = p_i \partial_\ell H_i + H_i \partial_\ell p_i$, giving $\partial_\ell(p_i H_i) = p_i(Y_i - nc_i - e_i \Psi_i / p_i - I_i) + r_i \tau_{j(\ell)} |H_i|$ ($I_A = 0$, $I_M = I$). Integrating over $[0, 1]$ under $H_i(0) = H_i(1) = 0$, summing over $i \in \{A, M\}$, and substituting (A.4) gives $\int_0^1 \sum_i p_i n c_i \, d\ell = \int_0^1 wn \, d\ell + b_t \bar{L}_t$. [← Back to Proposition 5] □

A.6 Proof of Proposition 6

A.6.1 Part 1: Firm attainment, active-research threshold, and pure-site sorting

Fix $\ell \in [0, 1]$ and sector $i \in \{A, M\}$ at bounded quotes $0 \leq p_i(\omega) \leq \bar{p} < \infty$, $\tau_{j(\ell)}(\omega) \geq 0$, and $w > 0$, and write $m_i(\phi) = \mathbb{E}_t[(p_i - r_i \tau_{j(\ell)})_+ G_i(\phi)]$.

LEMMA A.1: *Under (21)–(22) with $a_i > 1/(1 - \alpha_i) > 1$, $m_i(\phi)$ is continuous on $[0, 1]$ and satisfies*

$$\bar{m}_i := \sup_{0 \leq \phi \leq 1} m_i(\phi) \leq \bar{p} \sup_{0 \leq \phi \leq 1} \mathbb{E}[G_i(\phi)] = \bar{p} \, \frac{a_i}{a_i - 1} < \infty. \quad \text{(A.5)}$$

For each $\phi \in [0, 1)$, optimal production labor is unique and given by

$$L_i^*(\phi) = \left[\frac{\alpha_i Z_i m_i(\phi)}{(1 + \psi_i) w}\right]^{1/(1-\alpha_i)}, \quad \text{(A.6)}$$

and the profiled bid rent

$$R_i(\phi; w) = \Lambda_i w^{-\frac{\alpha_i}{1-\alpha_i}} m_i(\phi)^{1/(1-\alpha_i)} - \Psi_i(\phi; w), \qquad \Lambda_i = (1-\alpha_i) Z_i \left[\frac{\alpha_i Z_i}{1+\psi_i}\right]^{\alpha_i/(1-\alpha_i)}, \tag{A.7}$$

is finite and attained on $[0, 1)$.

Proof. Since $G_i(\phi; \ell, \omega) = 1 + \mathbf{1}_{\{U_i < \phi\}}(V_i - 1)$ with $U_i = \Phi(X_{it}^s(\ell))$ independent of $V_i = \Phi(X_{it}^g(\ell))^{-1/a_i}$ by construction and, a single site being atomistic, independent of the quoted (p_i, τ_j) at ℓ (a maintained restriction on the quote fields), $m_i(\phi) = m_i(0) + \phi\, \mathbb{E}_t[(p_i - r_i \tau_{j(\ell)})_+ (V_i - 1)] = m_i(0)[1 + \kappa_i(\ell)\phi]$ is affine on $[0, 1]$ with $\mathbb{E}[G_i(\phi)] = 1 + \phi/(a_i - 1) \le a_i/(a_i - 1) < \infty$, giving (A.5). Strict concavity in L^{α_i} yields (A.6)–(A.7), and $\Psi_i(\phi; w) \ge w\chi_i/(1-\phi) \to \infty$ as $\phi \uparrow 1$ ensures attainment on $[0, 1)$. □

At an interior optimum $\tilde{\phi}_i \in (0, 1)$, the envelope theorem gives

$$Z_i L_i^*(\tilde{\phi}_i)^{\alpha_i}\, \partial_\phi \mathbb{E}_t\Big[(p_i - r_i \tau_{j(\ell)})_+ G_i(\tilde{\phi}_i)\Big] = \partial_\phi \Psi_i(\tilde{\phi}_i; w) = \frac{w\chi_i}{(1-\tilde{\phi}_i)^2}, \tag{A.8}$$

matching (31). For $w_2 > w_1$ with optimal plans (ϕ_2, L_2) and (ϕ_1, L_1), writing $R_i(\phi; w, \ell)/w = \pi_i^0(w, \ell)[1 + \Gamma_i(\phi; \ell)] - C_i(\phi)$ with $C_i(\phi) = \Psi_i(\phi; w, \ell)/w$ and summing the revealed-preference inequalities gives

$$[\pi_i^0(w_2, \ell) - \pi_i^0(w_1, \ell)]\,[\Gamma_i(\phi_2; \ell) - \Gamma_i(\phi_1; \ell)] \ge 0. \tag{A.9}$$

Since $\pi_i^0(w_2, \ell) < \pi_i^0(w_1, \ell)$ and $\Gamma_i(\cdot; \ell)$ is strictly increasing, $\phi_2 \le \phi_1$, $m_i(\phi_2; \ell) \le m_i(\phi_1; \ell)$, and $L_2 = L_i^*(\phi_2; w_2, \ell) < L_i^*(\phi_1; w_1, \ell) = L_1$: thus $\phi_i^*(w, \ell)$ is weakly decreasing and $L_i^*(\phi_i^*(w, \ell); w, \ell)$ is strictly decreasing in w. Since $\partial^2_{\phi\phi}[m_i(\phi)^{1/(1-\alpha_i)}] = \frac{\alpha_i}{(1-\alpha_i)^2} m_i(\phi)^{(2\alpha_i - 1)/(1-\alpha_i)} [\partial_\phi m_i]^2$ with $\partial_\phi m_i = m_i(0; \ell)\kappa_i(\ell)$, strict concavity $\partial^2_{\phi\phi} R_i(\phi; w, \ell) < 0$ on $(0, 1)$ holds for $\chi_i > \bar{\chi}_{it} \equiv \frac{\alpha_i}{2(1-\alpha_i)^2} \sup_{\ell,\, w \in [\underline{w}_t, \overline{W}_t],\, \phi \in (0,1)} (1-\phi)^3 [m_i(0; \ell)\kappa_i(\ell)]^2 \Lambda_i(\ell) w^{-1/(1-\alpha_i)} m_i(\phi; \ell)^{(2\alpha_i - 1)/(1-\alpha_i)}$, so at most one interior stationary point $\tilde{\phi}_i(w, \ell)$ exists, and (32) uniquely selects $\phi_i^*(w, \ell) \in \{0, \tilde{\phi}_i(w, \ell)\}$. The reservation threshold is well defined: $S_i(\pi) \equiv \sup_{\phi \in (0,1)} [\pi \Gamma_i(\phi; \ell) - f_i(\ell) - \chi_i/(1-\phi)]$ is convex and nondecreasing in π as a supremum of affine functions with nonnegative slopes, $S_i(0) = -f_i(\ell) - \chi_i < 0$, and $S_i(\pi) \to \infty$ when $\kappa_i(\ell) > 0$, so S_i has a unique root $\underline{\pi}_i(\ell) \in (0, \infty)$ and research is active exactly when $\pi_i^0(w, \ell) > \underline{\pi}_i(\ell)$; at equality both 0 and $\tilde{\phi}_i$ are optimal and the convention $\phi_i^* = 0$ applies. If $\kappa_i(\ell) = 0$, research is never active ($\underline{\pi}_i(\ell) = \infty$), and if $m_i(0; \ell) = 0$ the sector is dormant at ℓ with $R_i = L_i^* = \phi_i^* = 0$. Under Condition 2, $R_A(\ell) - R_M(\ell)$ crosses zero transversally at a unique $b_{ct} \in (0, 1)$, assigning $[0, b_{ct}]$ to agriculture and $(b_{ct}, 1]$ to manufacturing. With e_{it} and ϕ_{it}^* fixed at every ℓ, (37) determines $A_{it}^+(\ell, \omega)$ pointwise, and Lemma S.C.1 (Online Appendix) ensures $F_{it}(\ell, \omega)$ in (38) is finite a.s. with finite mean everywhere on $[0, 1]$.

A.6.2 Part 2: Existence and norm-contraction global uniqueness

Proof. On $\mathcal{W}_t \times \mathcal{P}_t \times \mathcal{M}_t$ (Condition 1), $G_i(\phi_i^*; \ell, \omega) \ge 1$ implies $L_i^*(\phi_i^*; w, \ell) \ge L_i^*(0; w, \ell) \propto w^{-1/(1-\alpha_i)}$ ($L_T^* \propto w^{-1/(1-\alpha_T)}$), whereas $\Psi_i(\phi_i^*; w, \ell) = O(w)$ vanishes as $w \downarrow 0$; hence aggregate labor demand exceeds $\bar{L}_t$ at $\underline{w}_t > 0$ and falls below $\bar{L}_t$ at $\overline{W}_t < \infty$. Given predetermined capacities and Condition 1's subsistence, construction, and service slack, maximizing the strictly concave Negishi-weighted Stone–Geary log-surplus $\int_0^1 \vartheta(\ell, \omega) n(\ell) [\eta \log(c_A(\ell, \omega) - \gamma) + (1-\eta) \log c_M(\ell, \omega)]\, d\ell$ at weights $\vartheta(\ell, \omega) = w(\ell) + b_t(\omega) - \gamma p_A(\ell, \omega)$ subject to (68) and $D_j(\omega) \le K_j$ (50) yields unique KKT spot prices $\widetilde{p}_i(\cdot, \omega) \in W^{1,\infty}([0, 1])$ ($\widetilde{p}_A(1, \omega) = 1$) and county tolls $\widetilde{\tau}_j(\omega) \ge 0$. Since $|\partial_\ell \widetilde{p}_i| \le \kappa(T_t)\widetilde{p}_i + r_i \widetilde{\tau}_{j(\ell)}$ a.e., substituting $(\widetilde{p}(w), \widetilde{\tau}(w))$ into $\mathcal{T}_t^w$ in (73) defines a reduced wage operator $\widetilde{\mathcal{T}}_t^w : \mathcal{W}_t \to \mathcal{W}_t$

on $C([0,1])$ with equicontinuous image. By Arzelà–Ascoli and Schauder, $\mathcal{T}_t$ admits a fixed point $q^* = (w^*, p^*, \tau^*)$, which is unique under Condition 3 since $\|q_1 - q_0\| = \|\mathcal{T}_t(q_1) - \mathcal{T}_t(q_0)\| \le \bar{\rho}_t \|q_1 - q_0\|$ with $\bar{\rho}_t < 1$. □

A.6.3 Part 3: Constructive verification from primitives

Proof. Step 1 (Frictionless benchmark). When $\kappa_0 = r_A = r_M = m_I = 0$ and $f_i + \chi_i > \bar{f}_i \equiv \sup_{\ell,\phi\in(0,1)}(1-\phi)\pi_i^0(w_t^0, \ell)\Gamma_i(\phi;\ell)$, prices and wages equalize across space ($p_{At} \equiv 1$, $p_{Mt} \equiv p_{Mt}^0 > 0$, $\tau_{jt} \equiv 0$, $\phi_{it}^* \equiv 0$, $w_t \equiv w_t^0$). Opposite-signed log-knowledge slopes ($\partial_\ell \log A_{At} < 0 < \partial_\ell \log A_{Mt}$) make $\log(R_{At}(\ell)/R_{Mt}(\ell))$ strictly decreasing in ℓ, pinning down a unique cutoff $b_{ct}^0 \in (0,1)$ with $\partial_\ell[R_{At}(b_{ct}^0) - R_{Mt}(b_{ct}^0)] < 0$, while labor clearing and Stone–Geary expenditure shares determine $(w_t^0, p_{Mt}^0, b_{ct}^0)$ in closed form.

Step 2 (Uniform price and wage bounds). For any accumulated stock $T_t(\ell) = T_0(\ell) + s_{j(\ell),t}(\ell) \ge T_0(\ell)$, higher $s_{j(\ell),t}$ expands carrier capacity K_j and reduces route loss $\kappa_0 e^{-T_t}$, so county tolls are bounded by their initial-stock ceilings $\overline{\tau}_j(0)$ and bounded below in expectation by $\underline{\tau}_{jt} > 0$. Integrating (44) from $p_{At}(1,\omega) = 1$ across counties $j = J, \dots, 1$ bounds $p_{it}(\ell,\omega) \in [\underline{p}_i, \overline{p}_i]$ and $w(\ell) \in [\underline{w}_{\text{nat}}, \overline{W}_{\text{nat}}]$ uniformly across all stock paths $\{s_{jt}\}_{j=1}^J$, verifying Conditions 1 and 2.

Step 3 (Contraction under small frictions). Since b_{ct} and any active-research thresholds cross zero transversally, $\mathcal{T}_t$ is C^1 on $\mathcal{W}_t \times \mathcal{P}_t \times \mathcal{M}_t$ and contracts at the frictionless benchmark with modulus $\rho_0 < 1$ governed by diminishing returns to labor ($-1/(1-\alpha_i) < 0$) and Stone–Geary demand monotonicity. By continuity of $\mathcal{D}_t$, $\sup_q \|\mathcal{D}_t(q)\| \le \frac{1}{2}(1+\rho_0) < 1$ persists for small $(\kappa_0 e^{-T_0}, r_A, r_M, m_I)$ uniformly along every admissible infrastructure path, verifying Condition 3. [← Back to Proposition 6] □

A.7 Proof of Proposition 7

A.7.1 Part 1: Recursive county contraction and state transitions

Let $K_{\mathbf{x}}^\theta$ be the one-period forecasting kernel on Ω under (10)–(12), with stage value $Q_j(\Omega, \mathbf{x}; V) = R_j(\mathbf{x}; \Omega) + \beta(K_{\mathbf{x}}^\theta V_j)(\Omega)$. The weight $W(\Omega_t) = 1 + \sum_{i\in\{A,M\}} \|A_{it}\|_\infty + \sum_{j=1}^J \overline{T}_{jt}$ has finite one-step conditional mean $(K_{\mathbf{x}}^\theta W)(\Omega_t) \le \kappa W(\Omega_t)$ because $\sup_{\ell\in[0,1]} A_{i,t+1}(\ell) \le \sup_{\ell\in[0,1]} F_{it}(\ell)$ has finite conditional mean by Lemma S.C.1 (Online Appendix).

THEOREM A.1: *On a forecast-invariant state domain where each $\mathbf{x} \in \mathcal{X}_t$ has a unique measurable private equilibrium, let $W \ge 1$ be a measurable weight and $\mathcal{C}$ a closed class of county-value vectors equipped with the weighted C^1 route-stock norm $\|V\|_{W,1} = \max_j \sup_\Omega W(\Omega)^{-1}(|V_j(\Omega)| + \sum_{k=1}^J \overline{T}_k |\partial_{\overline{T}_k} V_j(\Omega)|)$. Suppose $|R_j(\mathbf{x};\Omega)| \le C_R W(\Omega)$ and $(K_{\mathbf{x}}^\theta W)(\Omega) \le \kappa W(\Omega)$. If for every $U, V \in \mathcal{C}$ the stage game has a unique measurable Nash profile $a_V(\Omega)$ satisfying $\|a_V(\Omega) - a_U(\Omega)\|_v \le L_a \|V - U\|_{W,1}$, the maximized payoff and its stock derivatives satisfy the Lipschitz bound with modulus $BW(\Omega)$ in $\|\cdot\|_v$, and the Nash Bellman operator $(\mathcal{B}^\theta V)_j(\Omega) = \max_{x_j} Q_j(\Omega, x_j, a_{V,-j}(\Omega); V)$ maps $\mathcal{C}$ into itself with $\beta\kappa + BL_a < 1$, then $\mathcal{B}^\theta$ has a unique fixed point $V^\theta \in \mathcal{C}$ and a unique Markov perfect policy a_{V^θ}.*

Proof. Holding actions fixed, $Q_j(\Omega, \mathbf{x}; V) - Q_j(\Omega, \mathbf{x}; U)$ and its stock derivatives are bounded by $\beta\kappa W(\Omega)\|V - U\|_{W,1}$, while shifting rival actions from $a_{V,-j}$ to $a_{U,-j}$ (and applying the envelope theorem for $a_{V,j}$) adds at most $BW(\Omega)L_a\|V - U\|_{W,1}$. Thus $\|\mathcal{B}^\theta V - \mathcal{B}^\theta U\|_{W,1} \le (\beta\kappa + BL_a)\|V - U\|_{W,1}$, yielding a unique fixed point $V^\theta \in \mathcal{C}$. □

COROLLARY A.1: *Under Theorem A.1, every realization of the physical shock and discovery laws from Ω_0 generates a unique path of allocations, knowledge frontiers, productivity gaps, and route stocks.*

Proof. Composing the unique conditional dated equilibrium map with the unique county policy a_{V^θ} fixes the allocation and investment profile at each state, and iterating (9), (37)–(39), and (53) determines the realized path by induction. □

A.7.2 Part 2: Constructive verification on the rescaled stock corridor

Proof. Fix the $J = 10$ counties $\mathcal{C}_j = [(j-1)/10, j/10]$ and the primitive regime of Proposition 6. At date t with $n \geq 1$ periods remaining in horizon H, county j maximizes $\Pi_{jt}^{(n),\theta}(\mathbf{x}_t; \Omega_t) = R_{jt}(\mathbf{x}_t; \Omega_t) + \beta \mathbb{E}_t^\theta[V_{j,t+1}^{(n-1),\theta}(\Omega_{t+1})]$ ($V_H^{(0),\theta} \equiv 0$), where $R_{jt} = \widetilde{R}_{jt} - m_I(x_{jt} + \frac{1}{2}\chi_I x_{jt}^2) D_{jt}^I$ with $D_{jt}^I \geq D_{\min} > 0$ from (60).

Step 1 (Stage-game diagonal dominance and attenuated cross-county sensitivity). Since $\mathbf{x}_t$ enters the current dated equilibrium only through $I_t(\ell; \mathbf{x}_t) = \sum_{r=1}^J e_{rt} k_{rt}(\ell) X_{rt}$ with $X_{rt} = m_I(x_{rt} + \frac{1}{2}\chi_I x_{rt}^2)$, the chain rule gives $\partial^2_{x_{jt}x_{jt}} \widetilde{R}_{jt} = (\partial_{X_{jt}} \widetilde{R}_{jt}) m_I \chi_I + (\partial^2_{X_{jt}X_{jt}} \widetilde{R}_{jt}) m_I^2 (1 + \chi_I x_{jt})^2$. When construction absorbs a small fraction of manufacturing capacity ($m_I = O(\varepsilon)$ with χ_I large so $|\partial_{X_{jt}} \widetilde{R}_{jt}| \leq \frac{1}{2} D_{jt}^I$), differentiating R_{jt} twice yields $\partial^2_{x_{jt}x_{jt}} R_{jt} \leq -\frac{1}{2} m_I \chi_I D_{jt}^I + O(m_I^2 (1 + \chi_I)^2)$ and $\partial^2_{x_{it}x_{jt}} R_{jt} = O(m_I^2(1 + \chi_I)^2)$ ($i \neq j$). When $H_{jt}^{(n-1)}(\overline{\mathbf{T}}_{t+1}; \mathbf{z}_t, \bar{\mathbf{z}}_t) = \mathbb{E}_t^\theta[V_{j,t+1}^{(n-1),\theta}]$ is concave in $\overline{T}_{j,t+1}$ with $|\partial^2_{\overline{T}_i \overline{T}_j} H_{jt}^{(n-1)}| \leq C_2 \varepsilon^2 / (\overline{T}_{i,t+1} \overline{T}_{j,t+1})^{(1+\alpha_T)/\alpha_T}$, the stage-$n$ pseudogradient $g_{jt}^{(n),\theta} = \partial_{x_j} \Pi_{jt}^{(n),\theta}$ is strongly monotone with modulus $\mu = \frac{1}{8} m_I \chi_I D_{\min} > 0$. By Proposition 4, $\mathbf{x}_t^{(n),\theta}$ is unique, and the implicit function theorem on each active interior set $\mathcal{A} = \{r : 0 < x_{rt}^{(n),\theta} < \bar{x}_{rt}\}$ yields the attenuated stock-sensitivity bound (74): $\|\partial_{\overline{T}_{it}} \mathbf{x}_t^{(n),\theta}\| \leq C_u \varepsilon / (\mu \min_r \overline{T}_{rt}^{(1+\alpha_T)/\alpha_T})$.

Step 2 (Backward induction on county value concavity). Differentiating $V_{jt}^{(n),\theta}(\Omega_t) = \Pi_{jt}^{(n),\theta}(\mathbf{x}_t^{(n),\theta}(\Omega_t); \Omega_t)$ with respect to $\overline{T}_{jt}$ via the envelope theorem eliminates $\partial_{x_j} \Pi_{jt}^{(n),\theta} \partial_{\overline{T}_j} x_{jt}^{(n),\theta} = 0$ a.e., leaving $\partial_{\overline{T}_j} R_{jt} + \beta \partial_{\overline{T}_j} H_{jt}^{(n-1)}$ plus rival feedback $\sum_{h \neq j} \partial_{x_h} \Pi_{jt}^{(n),\theta} \partial_{\overline{T}_j} x_{ht}^{(n),\theta}$. Because carrier and route-loss savings exhibit diminishing returns $\partial^2_{\overline{T}_j \overline{T}_j} R_{jt} \leq -c_R \varepsilon / \overline{T}_{jt}^{(1+2\alpha_T)/\alpha_T} < 0$ and expectations smooth rival policy kinks against the bounded-score truncated-Gaussian kernel of $\mathbf{z}_t$, (74) restricts the rival feedback derivative to $O((\varepsilon/\mu)\varepsilon^2 / (\overline{T}_{it} \overline{T}_{jt})^{(1+\alpha_T)/\alpha_T})$, preserving own-stock concavity and small cross-stock derivatives of $H_{j,t-1}^{(n)}$ for $n = 1, \ldots, H$.

Step 3 (Balanced-growth corridor and infinite-horizon convergence). Rescaling county stocks and increments by $g^t = 1.02^t$ via $\widetilde{T}_{jt} = g^{-t} \overline{T}_{jt}$ and $\bar{d}_j = g^{-t} d_{jt} = |\mathcal{C}_j|^{-1} \int_{\mathcal{C}_j} \iota_{j,1881}(\ell)\, d\ell$ gives $\widetilde{T}_{j,t+1} = g^{-1}(\widetilde{T}_{jt} + \bar{d}_j x_{jt})$. Discounted marginal carrier savings balance $m_I(1 + \chi_I x_{jt}) D_{jt}^I$ in the interior branch of (61) on a compact corridor $\mathcal{K} = \prod_{j=1}^J [L_j, U_j]$ at intensities $x_{jt}^{(n),\theta} \in [\underline{a}_j, \overline{a}_j] \subset (0, 1)$ satisfying $(g-1)L_j / \bar{d}_j \leq \underline{a}_j \leq \overline{a}_j \leq (g-1)U_j / \bar{d}_j$, which maps $\mathcal{K}$ into itself. Along $\mathcal{K}$, the state-weight drift is $\kappa = g^{1-\alpha_T}$ and $BL_a = O(\varepsilon/\mu)$, so $\beta\kappa + BL_a = \beta g^{1-\alpha_T} + O(\varepsilon/\mu) < 1$, and $(V_{jt}^{(H),\theta}, \mathbf{x}_t^{(H),\theta})$ contract geometrically as $H \to \infty$ to a unique infinite-horizon Markov perfect equilibrium. [← Back to Proposition 7] □

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